

Household behavior at the frozen calibration

2007 stationary reference — block0506, September 28 verified export

Common-primary reference loss: 19.581. All preferences, including the saved child-benefit parameter, remain fixed. The replacement-fertility normalization belongs to calibration and is not reapplied to shocks.

The frozen reference is authenticated against the original block0506 export: all 260 serialized parameter fields, the complete target and parameter tables, source identity and checkpoint identity are pinned in the compact reference manifest. The original 17 standard diagnostic plots are retained. This readout adds occupied-state views and one explicitly prescribed-price experiment; no calibration or measurement change is adopted.

The age-25 fertility miss is concentrated among mothers. Motherhood is 45.01% in the model versus 45.73% in the data, but capped children per mother are 1.190 versus 1.770. The conditional-count component accounts arithmetically for 96.14% of the early-fertility gap. This is a decomposition of a miss, not a causal finding or an impossibility result.

Explicit first births are most numerous in the youngest cell, while the first-birth probability among still-childless households peaks in the age-26 cell. Second-birth flows peak at age 30 and third-birth flows at age 38. Owners have higher conditional birth rates than renters at the same age: in the age-30 cell, first-birth probabilities are 39.69% and 21.80%, respectively. These probabilities cover a four-year cell and retain wealth/income selection within tenure groups.

Realized mean housing rises from 6.10 rooms at age 30 to 6.77 at age 42, then falls. Ownership rises from 51.34% at age 30 to 83.32% at age 66 and 94.68% at age 82. Beginning net financial wealth peaks at 5.31 at age 66 and reaches -1.24 at age 82, while beginning net worth remains positive at 2.34. Financial wealth and gross house value are separately observed; mortgage balances and housing equity are not separately identified.

Native saving limits bind for 25.11% of occupied renters, 4.72% of buyers and 4.54% of owner stayers. In the age-30 cell, the renter share is 42.80%. These are weighted boundary incidences under the saved discrete-grid solution, not shadow values. They exclude the incidence of households prevented from buying; a low buyer binding share cannot establish that down payments are unimportant.

Both fresh reference controls exactly reproduce 113 numerical arrays, all 14 fit rows, all 31 reported parameters and all 17 standard PNGs. The single permanent 10% price/rent shock passes the household and accounting gates with unchanged preferences. Immediate birth flows fall 4.19%, with 87.10% of that fall coming from first births. Separately recomputed cohort completed fertility falls from 2.100 to 1.988, and mean first-birth age rises from 25.933 to 26.220. Fertility is not renormalized.

Interpretation and limits

The price experiment is partial equilibrium. It leaves housing excess supply of 0.545 rooms per reference household on impact and 0.757 in the normalized cohort calculation. The cohort adult-entry replacement gap is 5.33%; that distribution is not a renewed demographic steady state. The tax and pension settings remain fixed and their applicable accounting checks pass.

Nonhousing consumption rises on impact but falls in the separately recomputed cohort. Initial owners retain their houses as prices rise, so housing revaluation is a plausible mechanism; this aggregate comparison does not identify that channel by itself. Consumption and saving integrate separate buyer/renter and owner-stayer policies.

Within the central occupied wealth support, childless-renter ownership probabilities generally increase with initial wealth, and birth-outcome housing choices are larger after a first birth. These conditional curves average selected income states; they are not causal birth effects. Cross-sectional plots against post-transaction financial wealth mix tenure selection with borrowing. High-wealth declines, local kinks and late-retirement borrowing remain questions for diagnosis, not certified mechanisms.

The separate chat Prepare transitions for frozen block0506 owns historical September 10-14 reconstruction, transition-code preparation and the requested credit/fixed-stock cases. Credit liberalization removes artificial limits while retaining lifetime solvency and repayment. The fixed-housing case freezes physical supply while prices and rents clear. A new steady state and the 10% supply expansion require a reconciled demographic, fiscal and estate closure; no such result is claimed here. Historical 2007-to-2023 re-estimation remains separate and deferred.

Torch jobs 18738188 and 18738587 extracted the saved state without solves. Job 18738157 completed exactly three lifecycle solves in 376.76 seconds within its 20-minute cap. The original checkpoint and all 30 authenticated export files remain unchanged. Nine new tiny-tail diagnostic logs followed an inherited logging path; they were moved, with unchanged hashes, into the corresponding experiment cases. The first extraction attempt failed on float32 probability normalization; the correction matched the native float64 convention and passed the unchanged tolerance.

Prescribed housing prices: impact and cohort composition

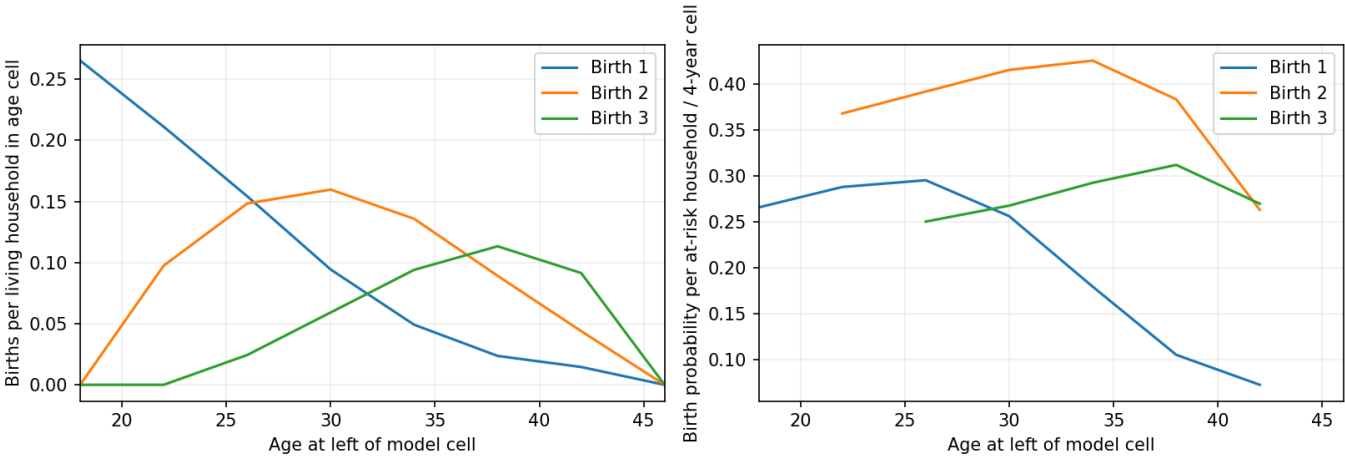
Only the asset price and implied rent rise by 10%. All preferences, earnings, taxes, pensions, entry endowments, survival, credit rules and supply primitives stay fixed. The impact column uses the exact reference pre-choice distribution and household policies for permanently higher prices. The cohort column separately recomputes composition with normalized entry. Neither is a cleared equilibrium or a transition.

Object	Reference	Impact	Cohort
Births per household / four-year period	0.11542	0.11059	0.10950
First births per household / four-year period	0.05016	0.04596	0.04813
Owners (% of all households)	66.82	65.85	64.64
Realized rooms per household	5.848	5.664	5.453
Nonhousing consumption per household	2.3414	2.3830	2.3177
Completed fertility	2.100	-	1.988
Asset price	0.78987	0.86886	0.86886
Implied rent	0.14240	0.15664	0.15664

Birth flows refer to four-year model periods. Completed fertility is undefined for the one-date impact column. Housing supply follows the unchanged supply curve when reporting excess demand; physical stock is not fixed in this diagnostic. The separately requested fixed-physical-stock transition is still being prepared. The full 14-row target comparison and 31-row parameter table follow the supplemental figures. Counterfactual fit against old data is a comparison, not a recalibration objective.

Supplemental: First and subsequent births

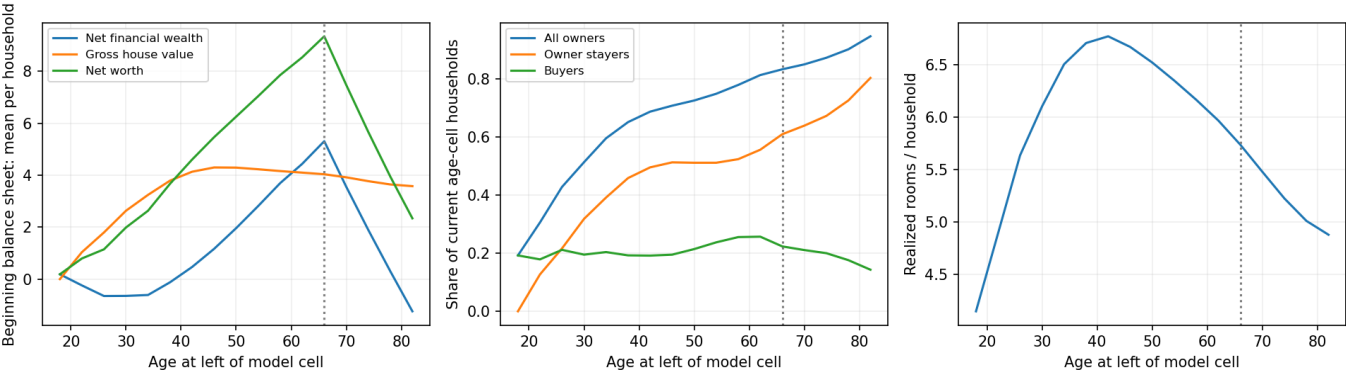
Supplemental: 2007 stationary reference — block0506, September 28 verified export



Left: explicit births per living household in each four-year age cell. Right: successful birth probability among households eligible for that birth. The denominator matters: an early peak in total first births differs from a peak in the probability among the remaining childless. Third births are explicit entries into the top child-count bin; demographic top-bin weights are not applied.

Supplemental: Realized lifecycle housing and balance sheets

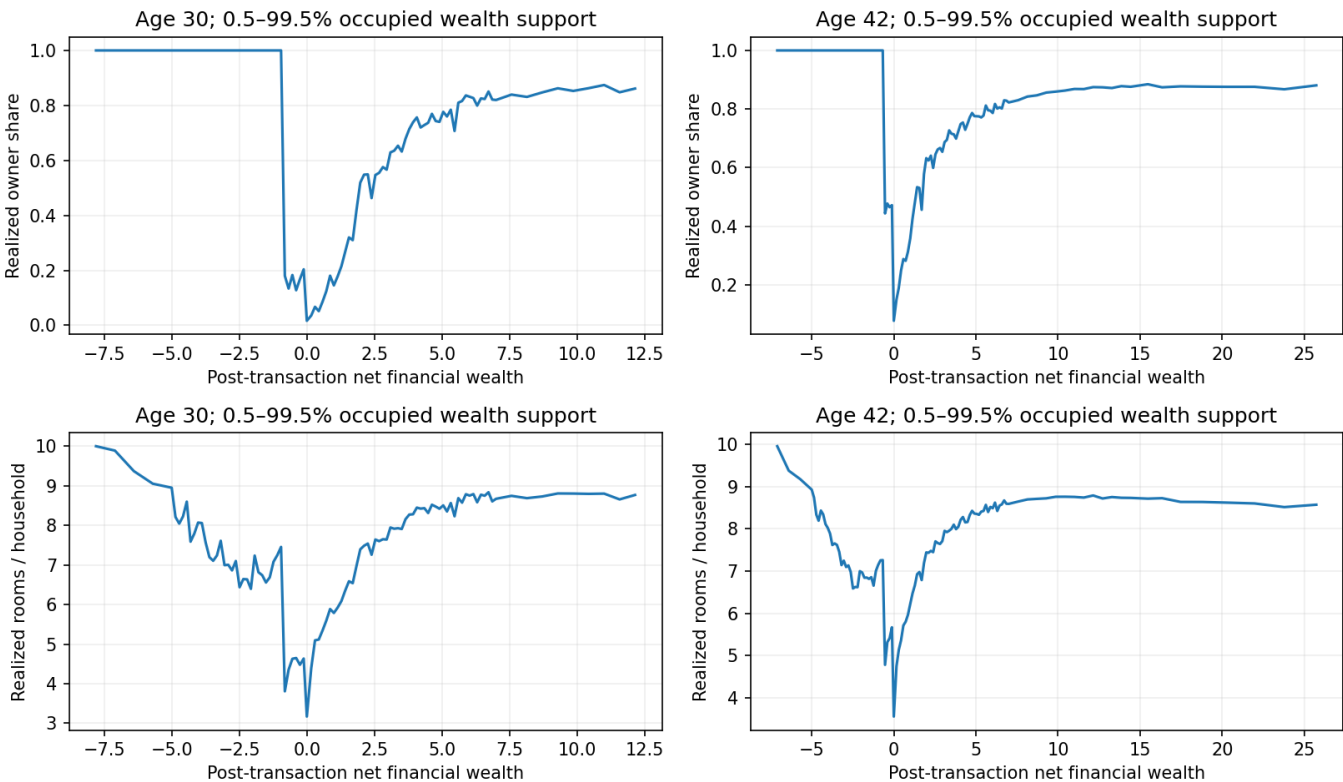
Supplemental: 2007 stationary reference — block0506, September 28 verified export



Beginning net worth equals net financial wealth plus gross house value before this period's tenure transaction, matching the authoritative wealth observer. The ownership and room panels show current realized choices. Buyers include previous owners moving to another owner product. The dotted line marks retirement at the age-66 model cell; these are stationary age profiles, not a historical transition.

Supplemental: Occupied wealth support

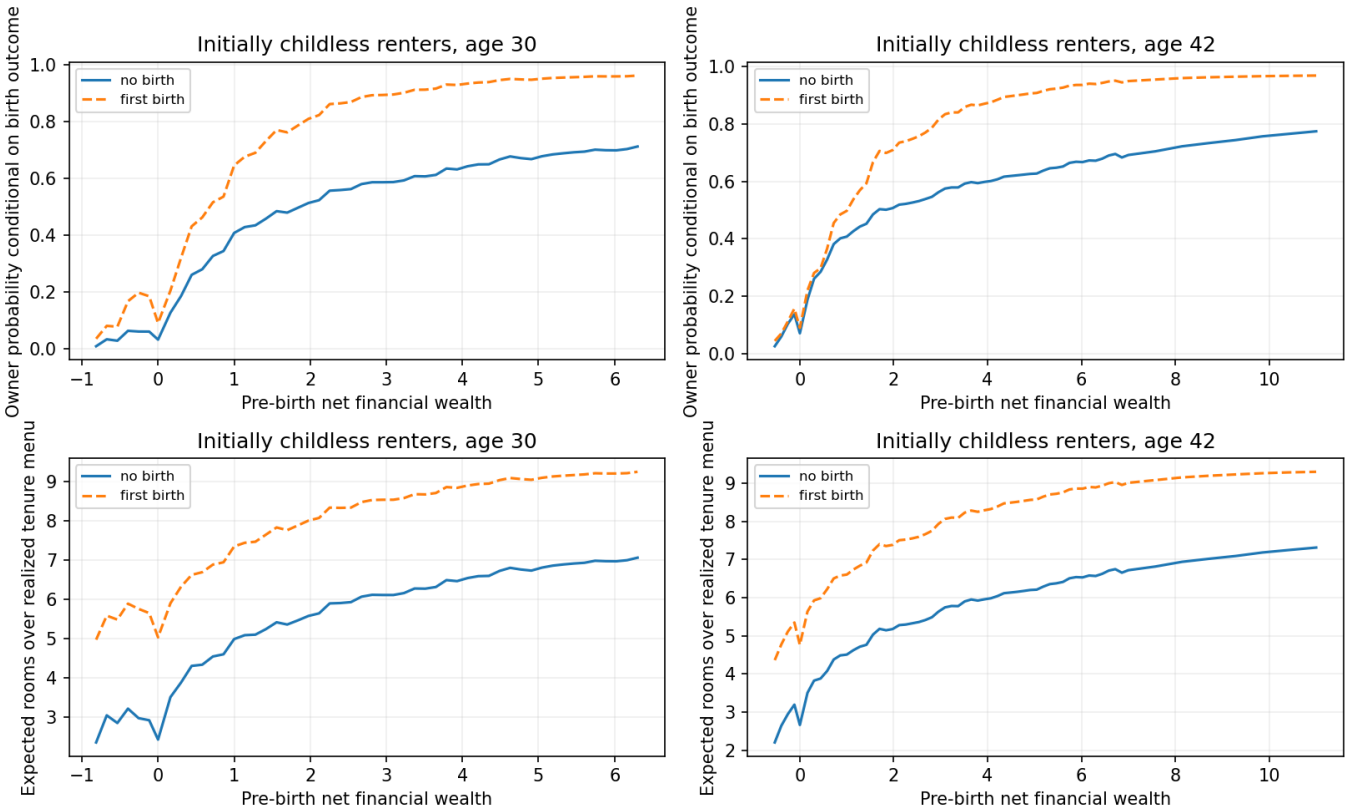
Supplemental: 2007 stationary reference — block0506, September 28 verified export



Conditional realized outcomes by post-transaction net financial wealth, restricted to the central 99% of each age cell's occupied wealth distribution. Negative financial wealth is heavily selected toward owners because they hold housing and debt. The resulting shape is not a causal wealth response and should not be read as a policy function.

Supplemental: Conditional choices of initially childless renters

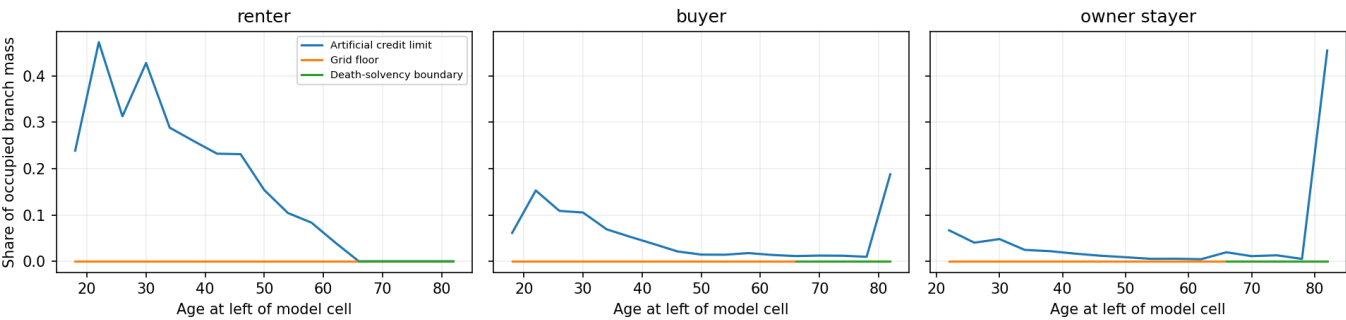
Supplemental: 2007 stationary reference — block0506, September 28 verified export



Expected ownership and rooms integrate actual tenure-choice probabilities and transaction interpolation within each birth-outcome branch. The initially occupied renter distribution supplies income weights. First-birth and no-birth branches therefore have different selected income compositions. These supplemental views do not replace the full-grid standard panels.

Supplemental: Occupied credit-limit incidence

Supplemental: 2007 stationary reference — block0506, September 28 verified export
Boundary incidence; death boundary is not a separate native constraint for renters/buyers



Each denominator is the actual renter, buyer or owner-stayer mass of that age. Saving is compared with the native credit floor, grid floor and, where death is possible, the net-estate solvency boundary. Constraint components may overlap and must not be added. Equality within 1e-8 asset units is a boundary incidence, not a shadow-price estimate. Purchase exclusion incidence is not measured.

Complete reference target fit

Normalization, scored moments and zero-weight validation rows retain their original definitions. CSVs retain full precision; gaps equal model minus target. These are 2007-reference comparisons.

Moment / role	Target	Model	Gap	Weight	Loss
initial_normalization / normalization	2.100	2.100	-1.663e-06	-	-
cps_childlessness / scored	0.198	0.201	0.003	3.553e+04	0.275
cps_exactly_one / scored	0.214	0.209	-0.004	2.695e+04	0.483
nchs_mean_age / scored	25.976	25.933	-0.043	139.828	0.264
nchs_share30 / validation	0.249	0.224	-0.026	0.000	0.000
wealth_earnings / scored	6.927	6.326	-0.600	7.595	2.737
bequest_wealth / scored	0.007	0.007	-2.336e-04	5.165e+06	0.282
old_dispersion / validation	3.516	3.069	-0.447	0.000	0.000
mean_rooms / scored	5.729	5.848	0.119	128.021	1.798
ownership_30_55 / scored	0.676	0.655	-0.021	2339.362	1.076
first_birth_rooms / scored	1.465	1.622	0.157	137.565	3.395
family_rooms / validation	0.385	0.353	-0.032	0.000	0.000
recent_parent_ownership / scored	0.128	0.120	-0.008	2.706e+04	1.758
early_fertility / scored	0.810	0.535	-0.274	100.000	7.513

Complete reference parameters and restrictions

Parameter	Estimate	Lower	Upper	Near bound	Role / restriction
H0	6.294	0.200	80.000	False	free in evening DUE calibration
beta_annual	0.963	0.940	0.990	False	free in evening DUE calibration
chi	1.094	0.100	5.000	False	free in evening DUE calibration
first_birth_fixed_cost	0.621	0.000	8.000	False	free in evening DUE calibration
kappa_fert	0.176	0.020	50.000	True	free in evening DUE calibration
kappa_fert_continuation	0.332	0.020	50.000	True	free in evening DUE calibration
theta0	0.125	0.000	8.000	False	free in evening DUE calibration
delta_alpha_jump	0.135	0.000	0.250	False	free in evening DUE calibration
child_benefit_curvature	0.061	0.000	0.800	False	free in evening DUE calibration
tenure_choice_kappa	0.012	0.001	0.100	False	free in evening DUE calibration
psi_child	0.136	-	-		normalized to completed fertility 2.1
child_benefit_CRRA_coefficient	0.127	-	-		derived from normalized one-child benefit
theta1	0.008	-	-		fixed external restriction
sigma	2.000	-	-		fixed
alpha_cons	0.733	-	-		fixed CEX childless expenditure share
delta_alpha	0.000	-	-		fixed zero later-child loading
h_P	0.000	-	-		no housing floor
utility_reference_rent	0.110	-	-		fixed substantive utility normalization
q_annual	0.020	-	-		author-retained 2% annual real rate
financed_share	0.800	-	-		inherited credit contract
housing_supply_elasticity	0.630	-	-		fixed provisional external mapping
payroll_tax	0.080	-	-		derived from adopted pension ratio
pension_period	0.918	-	-		balanced PAYGO
annual_depreciation	0.014	-	-		adopted
period_depreciation	0.055	-	-		compounded
annual_property_tax	0.011	-	-		adopted
period_property_tax	0.042	-	-		linear period convention
selling_cost	0.060	-	-		retained
rental_cap	6.000	-	-		retained provisional
wealth_grid_nodes	160.000	-	-		retained exact grid
income_states	15.000	-	-		retained B15

Experimental +10% prescribed price and rent

Complete target comparison for the normalized-entry cohort at permanently higher housing prices; all 31 reference parameters remain fixed. The unchanged 2007 targets provide a comparison, not an objective for re-estimation. The old replacement benchmark is displayed but not imposed. Common-primary comparison loss is 99.333; its increase is not evidence of a failed counterfactual.

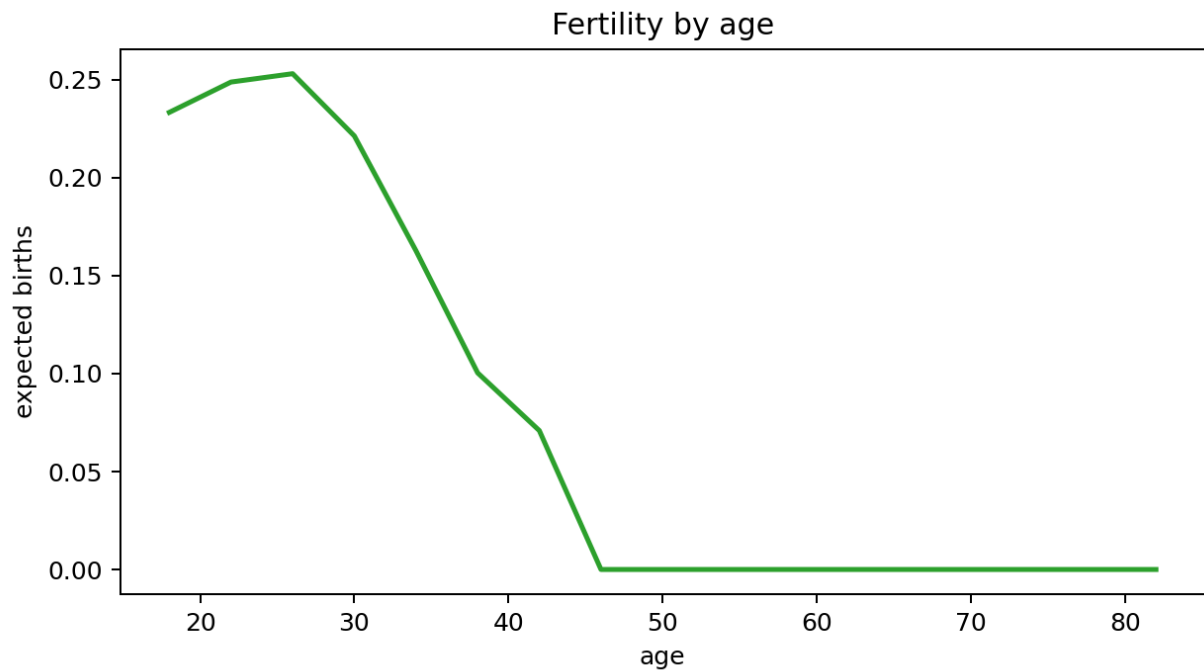
Moment / role	Target	Model	Gap	Weight	Loss
initial_normalization / reference replacement benchmark; not imposed after shock	2.100	1.988	-0.112	-	-
cps_childlessness / scored	0.198	0.235	0.037	3.553e+04	49.010
cps_exactly_one / scored	0.214	0.221	0.008	2.695e+04	1.557
nchs_mean_age / scored	25.976	26.220	0.244	139.828	8.315
nchs_share30 / validation	0.249	0.240	-0.009	0.000	0.000
wealth_earnings / scored	6.927	6.292	-0.635	7.595	3.059
bequest_wealth / scored	0.007	0.007	-2.461e-04	5.165e+06	0.313
old_dispersion / validation	3.516	3.088	-0.428	0.000	0.000
mean_rooms / scored	5.729	5.453	-0.276	128.021	9.771
ownership_30_55 / scored	0.676	0.626	-0.050	2339.362	5.944
first_birth_rooms / scored	1.465	1.636	0.171	137.565	4.010
family_rooms / validation	0.385	0.388	0.003	0.000	0.000
recent_parent_ownership / scored	0.128	0.111	-0.017	2.706e+04	7.410
early_fertility / scored	0.810	0.494	-0.315	100.000	9.943

Experimental +10% prescribed price and rent: complete parameters

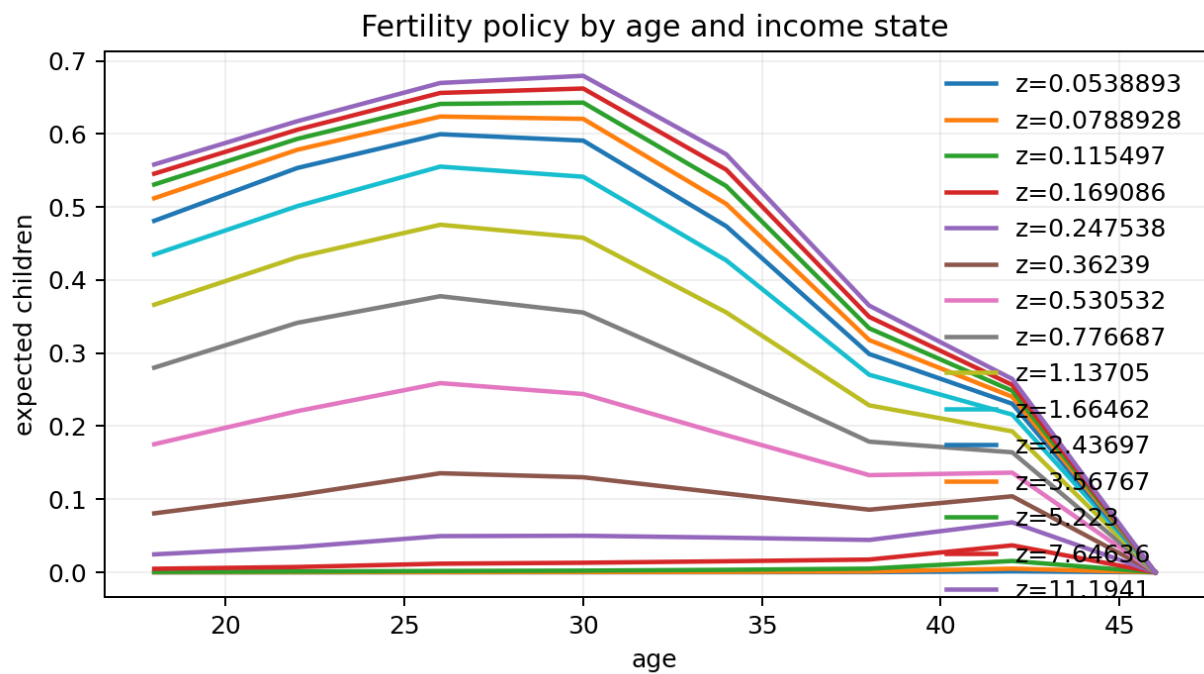
Parameter	Estimate	Lower	Upper	Near bound	Role / restriction
H0	6.294	0.200	80.000	False	Fixed inherited reference value; no estimation or fertility renormalization
beta_annual	0.963	0.940	0.990	False	Fixed inherited reference value; no estimation or fertility renormalization
chi	1.094	0.100	5.000	False	Fixed inherited reference value; no estimation or fertility renormalization
first_birth_fixed_cost	0.621	0.000	8.000	False	Fixed inherited reference value; no estimation or fertility renormalization
kappa_fert	0.176	0.020	50.000	True	Fixed inherited reference value; no estimation or fertility renormalization
kappa_fert_continuation	0.332	0.020	50.000	True	Fixed inherited reference value; no estimation or fertility renormalization
theta0	0.125	0.000	8.000	False	Fixed inherited reference value; no estimation or fertility renormalization
delta_alpha_jump	0.135	0.000	0.250	False	Fixed inherited reference value; no estimation or fertility renormalization
child_benefit_curvature	0.061	0.000	0.800	False	Fixed inherited reference value; no estimation or fertility renormalization
tenure_choice_kappa	0.012	0.001	0.100	False	Fixed inherited reference value; no estimation or fertility renormalization
psi_child	0.136	-	-		Fixed inherited reference value; no estimation or fertility renormalization
child_benefit_CRRA_coefficient	0.127	-	-		Fixed inherited reference value; no estimation or fertility renormalization
theta1	0.008	-	-		Fixed inherited reference value; no estimation or fertility renormalization
sigma	2.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization
alpha_cons	0.733	-	-		Fixed inherited reference value; no estimation or fertility renormalization
delta_alpha	0.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization
h_P	0.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization
utility_reference_rent	0.110	-	-		Fixed inherited reference value; no estimation or fertility renormalization
q_annual	0.020	-	-		Fixed inherited reference value; no estimation or fertility renormalization
financed_share	0.800	-	-		Fixed inherited reference value; no estimation or fertility renormalization
housing_supply_elasticity	0.630	-	-		Fixed inherited reference value; no estimation or fertility renormalization
payroll_tax	0.080	-	-		Fixed inherited reference value; no estimation or fertility renormalization
pension_period	0.918	-	-		Fixed inherited reference value; no estimation or fertility renormalization

Parameter	Estimate	Lower	Upper	Near bound	Role / restriction
annual_depreciation	0.014	-	-		Fixed inherited reference value; no estimation or fertility renormalization
period_depreciation	0.055	-	-		Fixed inherited reference value; no estimation or fertility renormalization
annual_property_tax	0.011	-	-		Fixed inherited reference value; no estimation or fertility renormalization
period_property_tax	0.042	-	-		Fixed inherited reference value; no estimation or fertility renormalization
selling_cost	0.060	-	-		Fixed inherited reference value; no estimation or fertility renormalization
rental_cap	6.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization
wealth_grid_nodes	160.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization
income_states	15.000	-	-		Fixed inherited reference value; no estimation or fertility renormalization

Reference: standard diagnostics

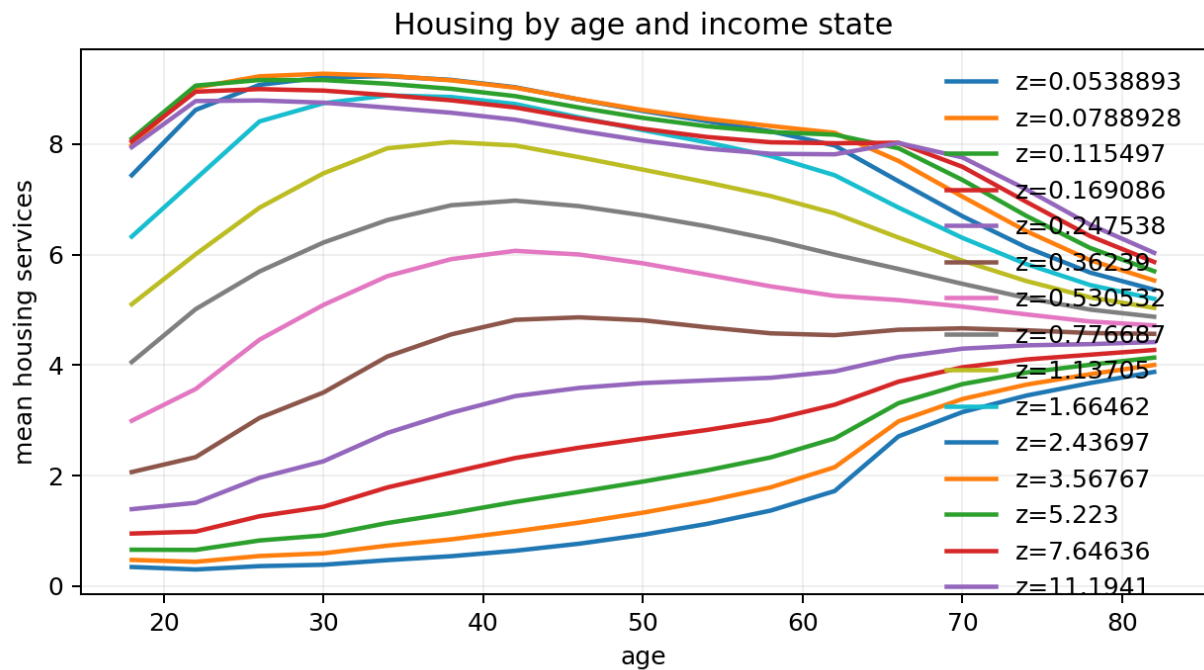


fertility by age

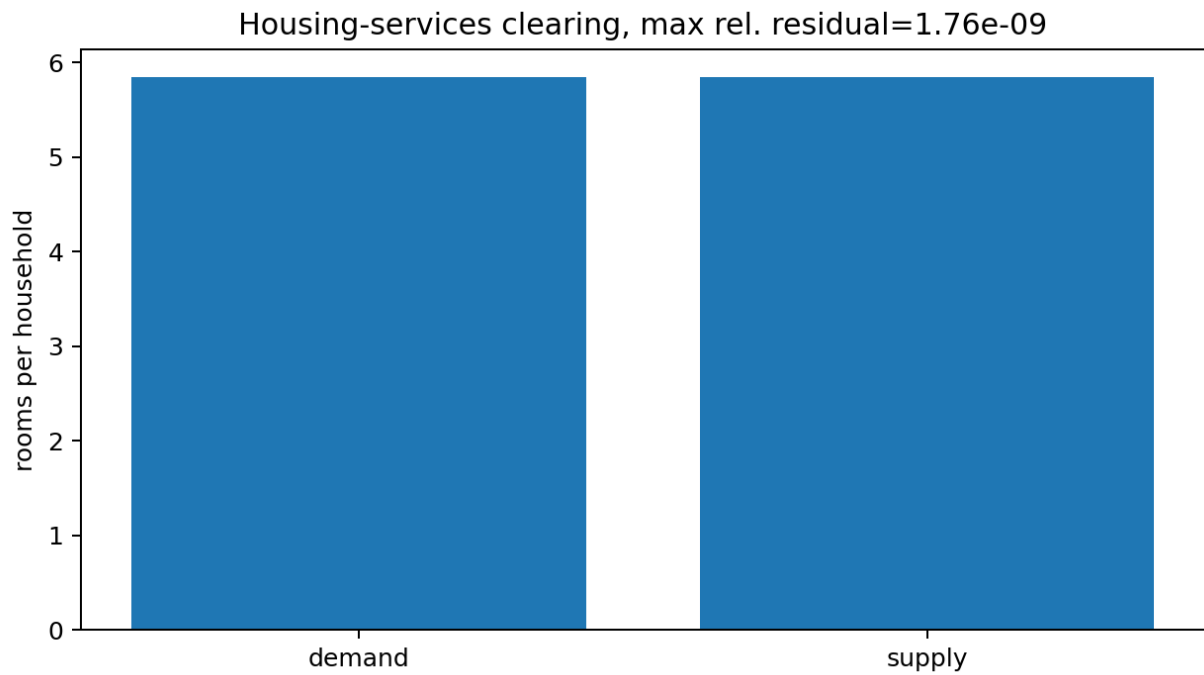


fertility policy by age income state

Reference: standard diagnostics



housing by age income state

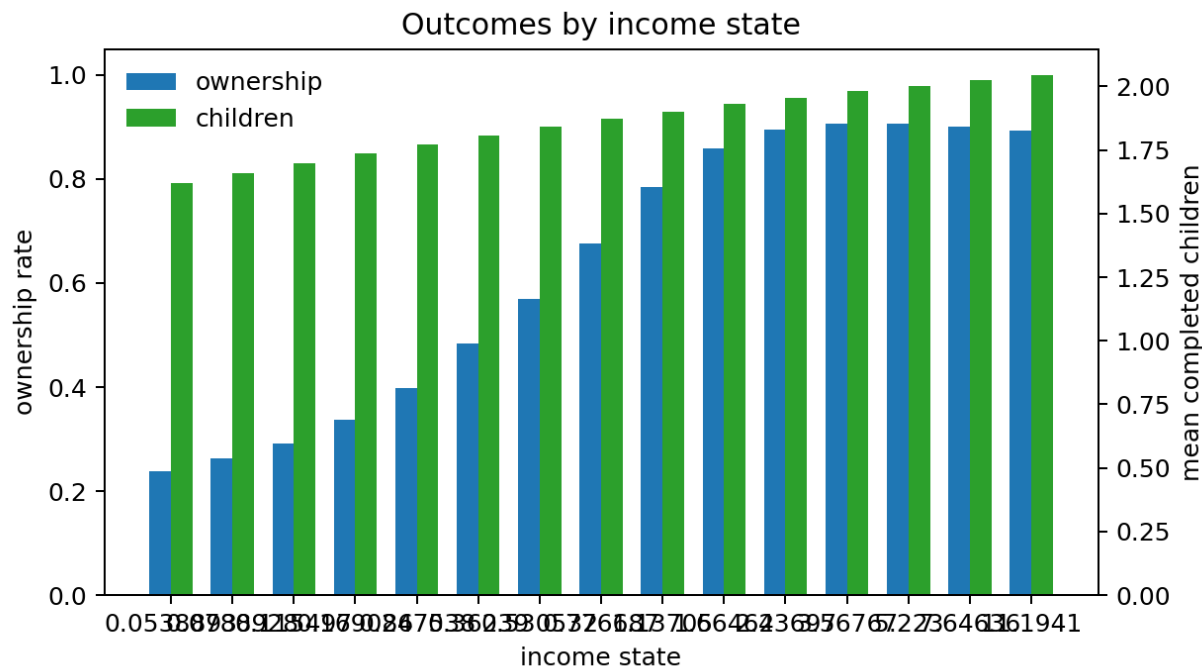


housing market

Reference: standard diagnostics

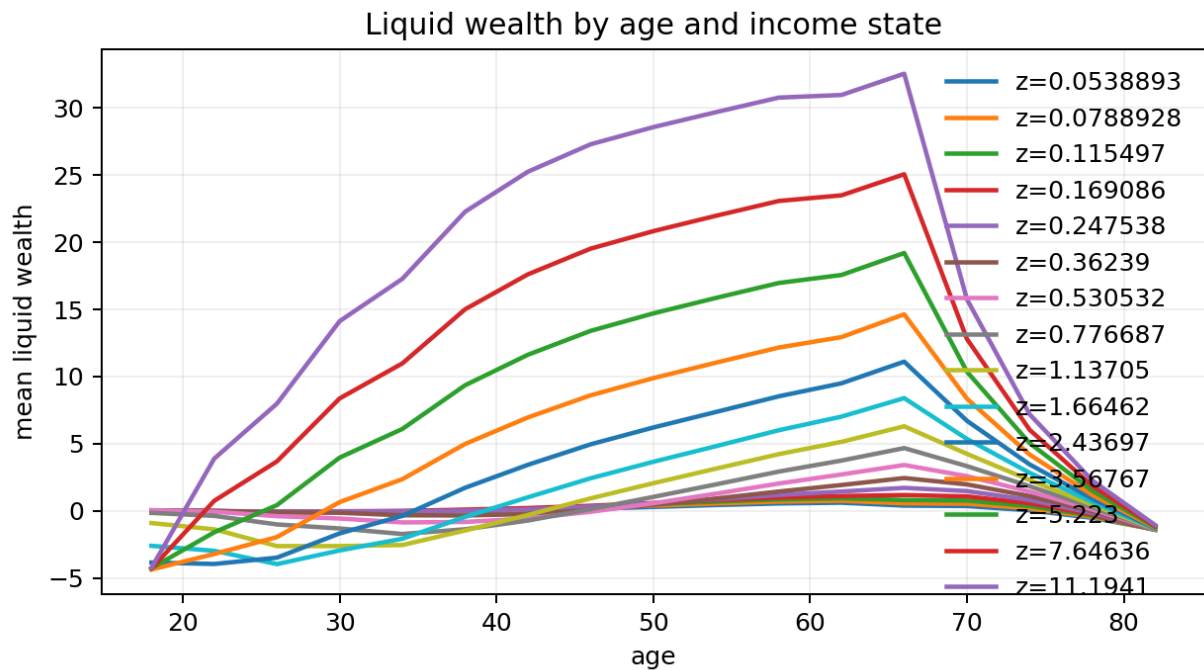


housing prices

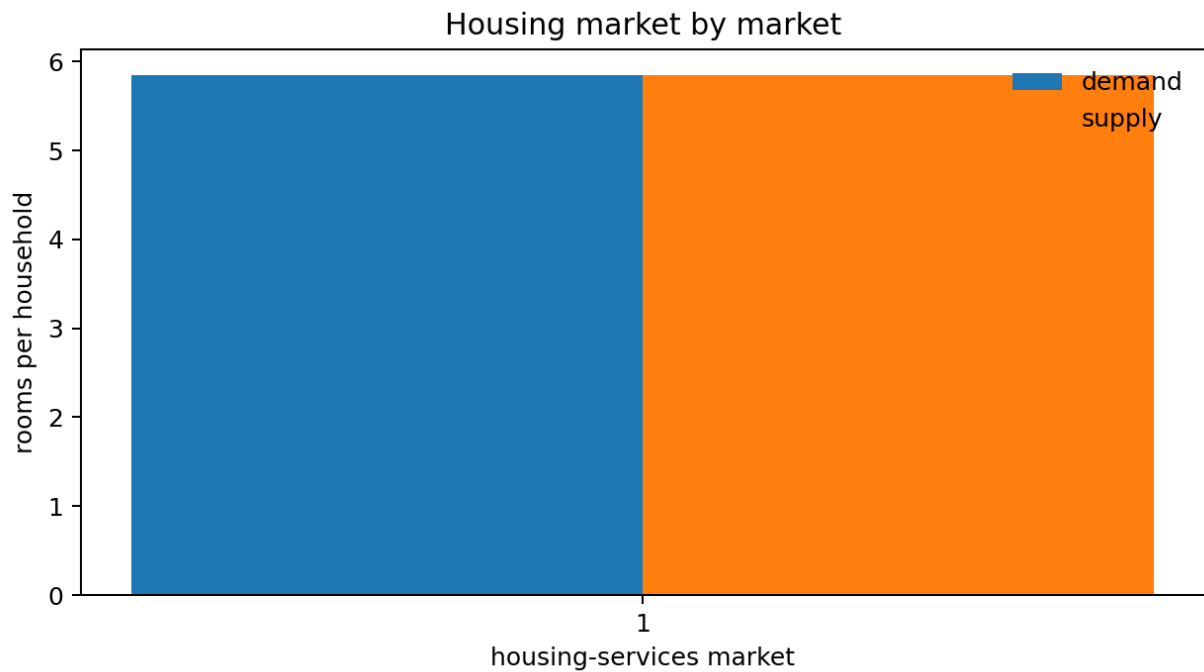


income state outcomes

Reference: standard diagnostics

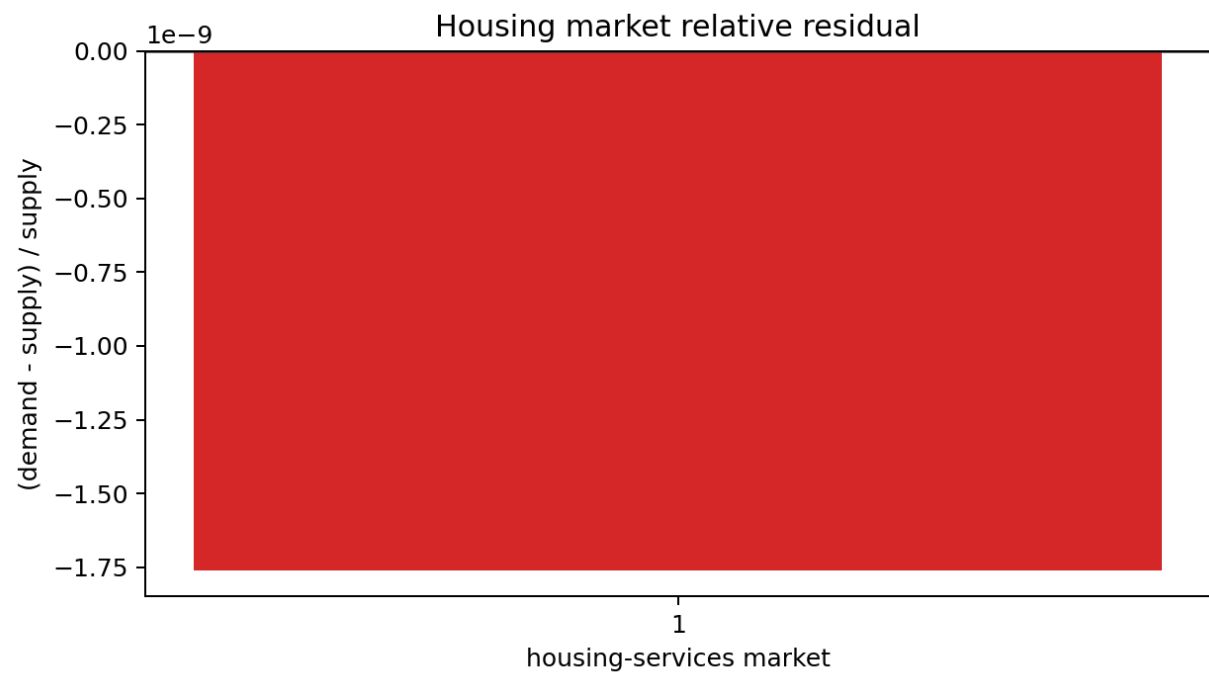


liquid wealth by age income state

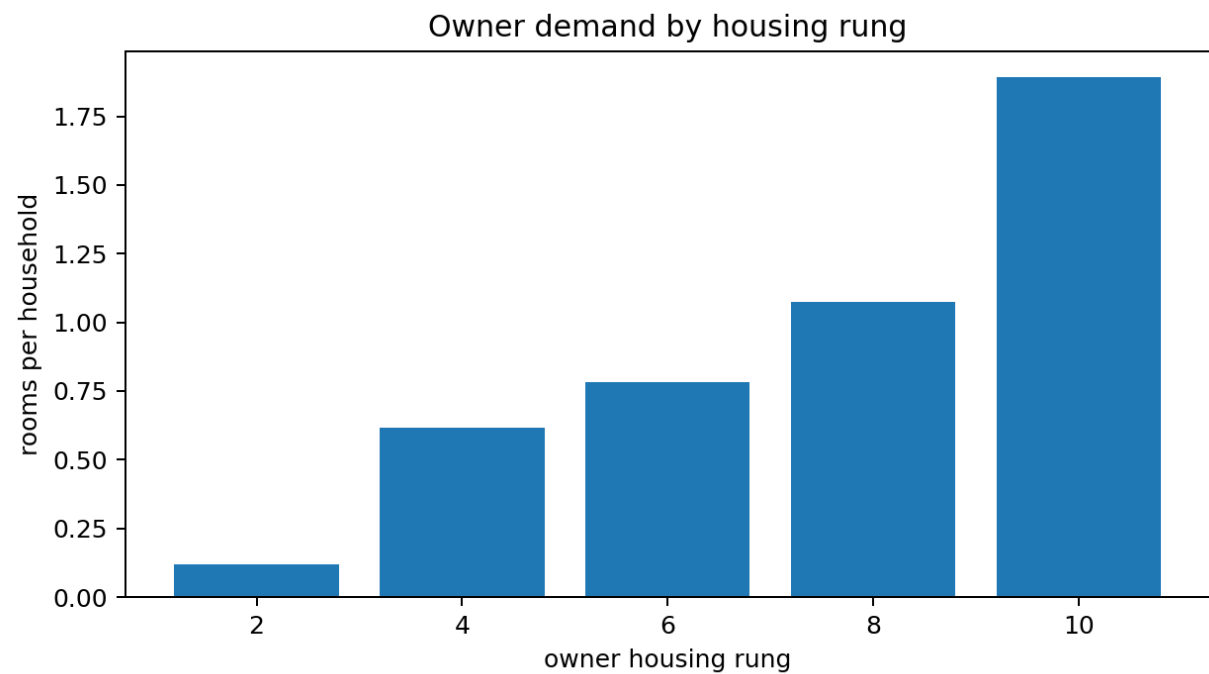


market clearing by market

Reference: standard diagnostics

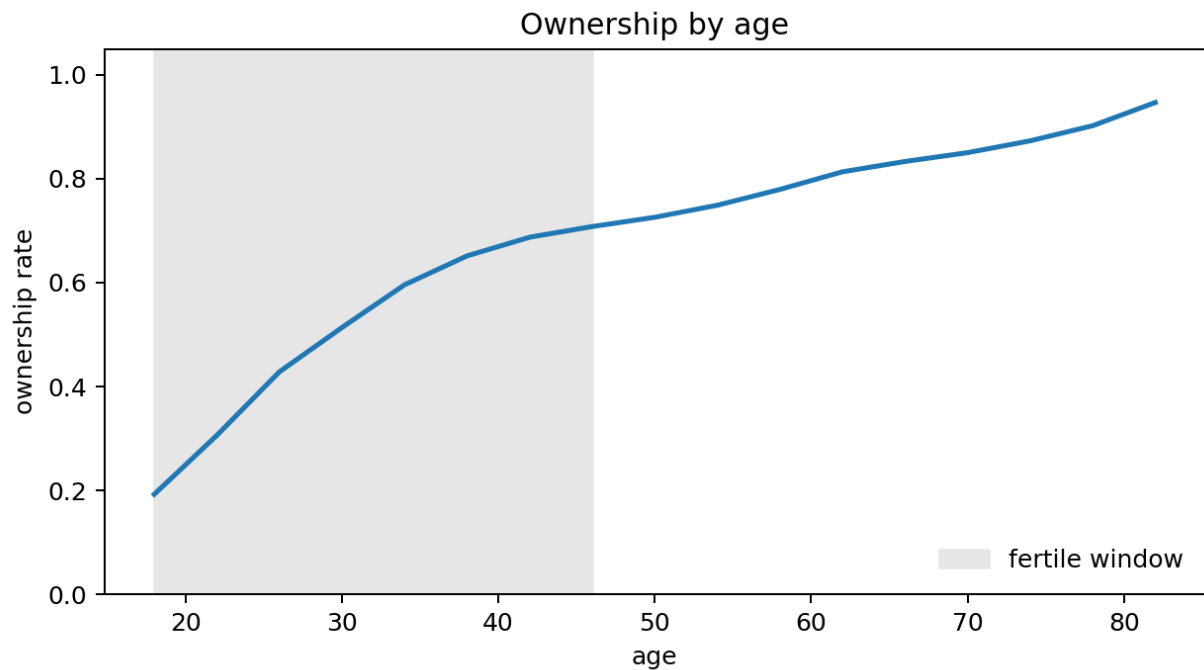


market clearing residuals

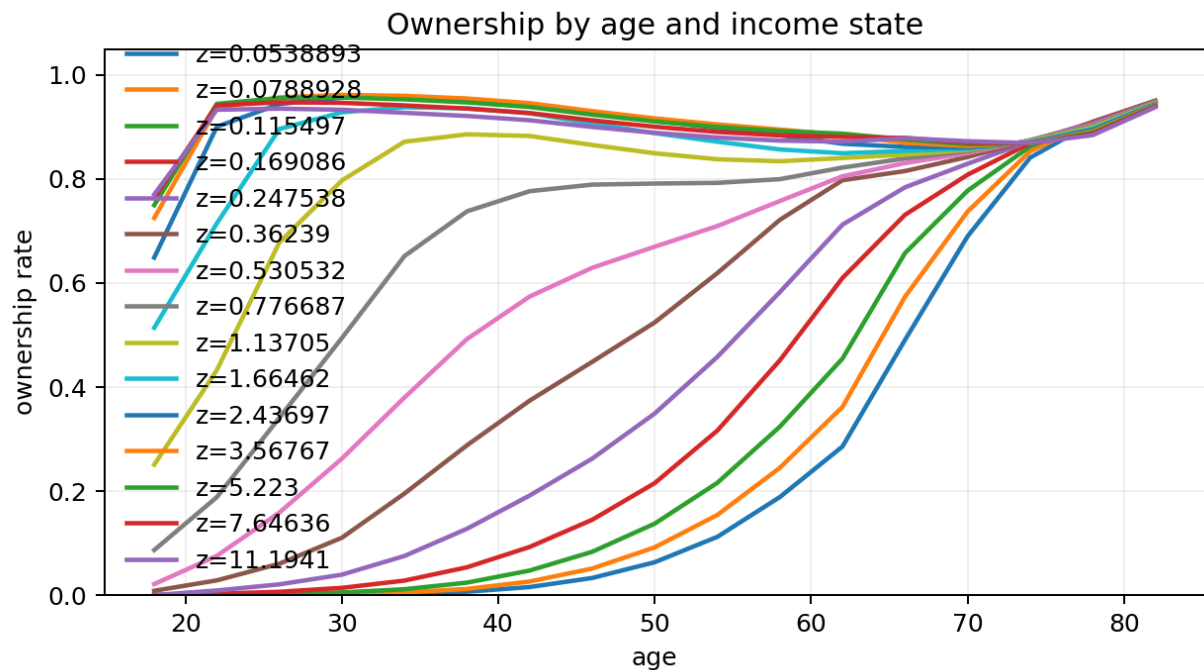


owner rungs

Reference: standard diagnostics

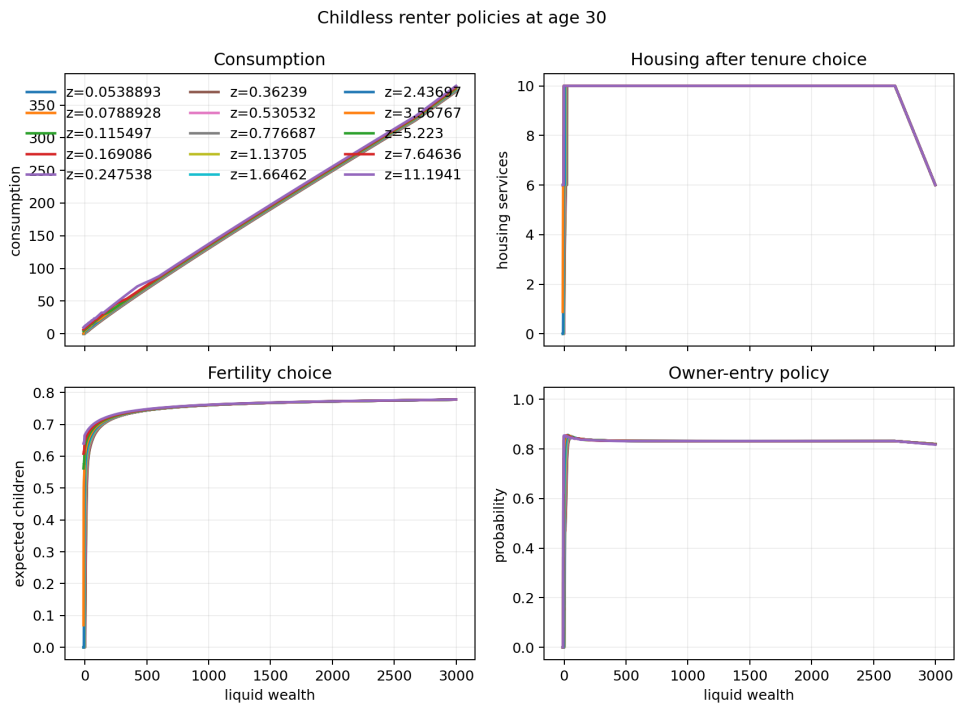


ownership by age

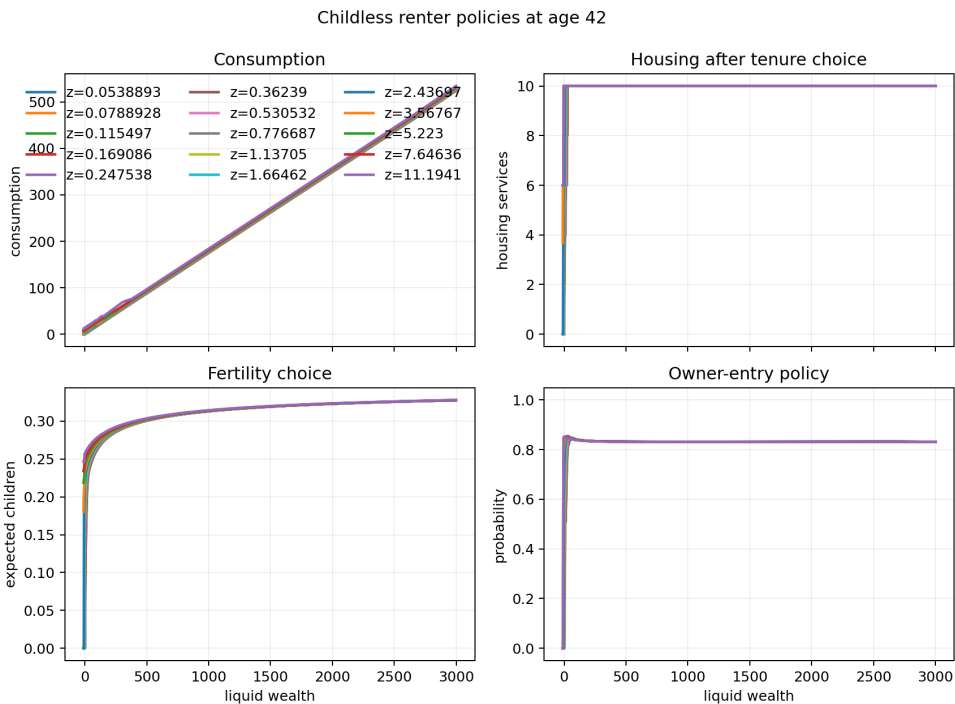


ownership by age income state

Reference: standard diagnostics

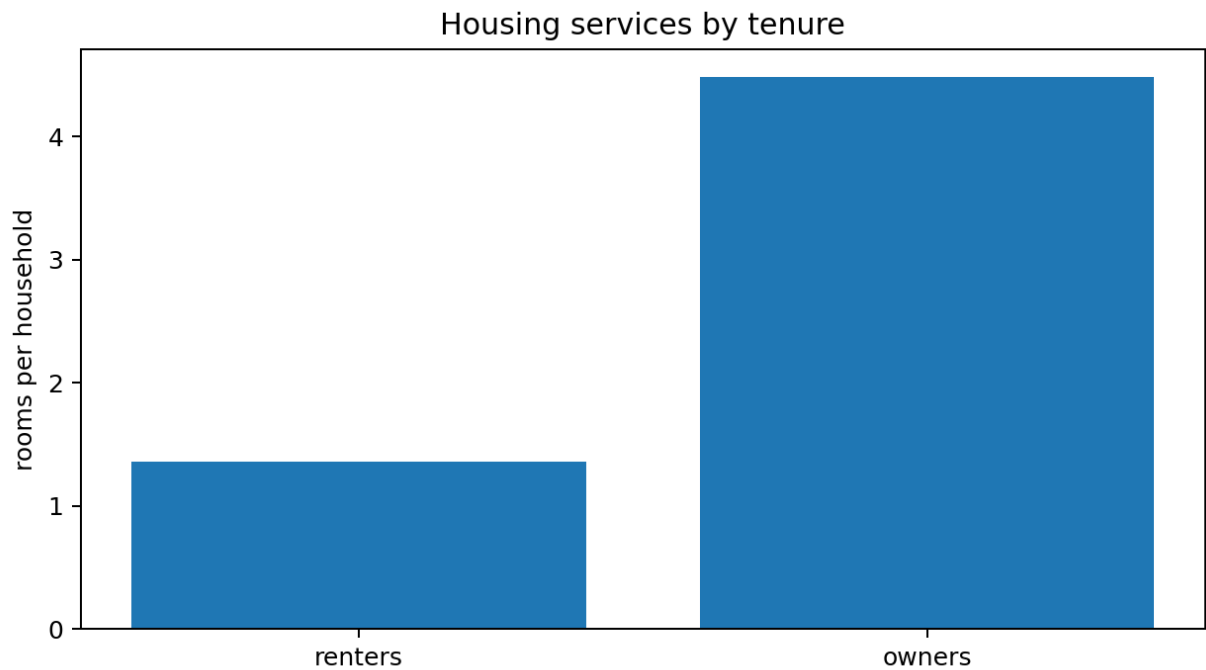


policy childless renter age30

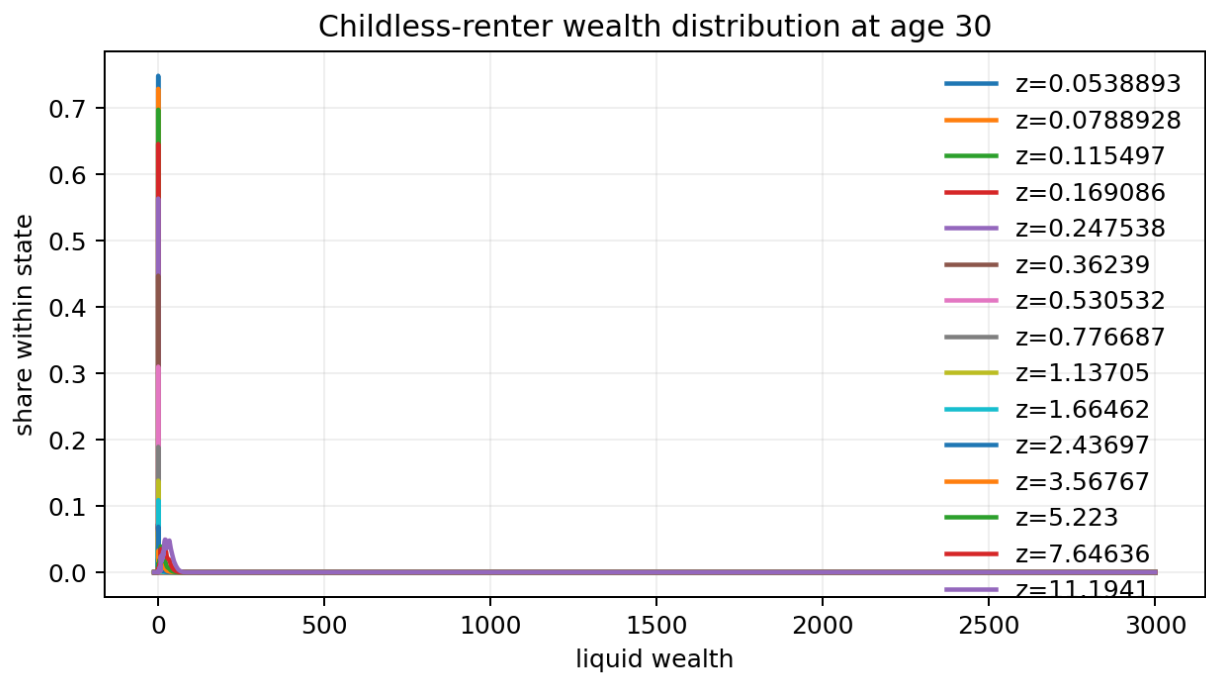


policy childless renter age42

Reference: standard diagnostics

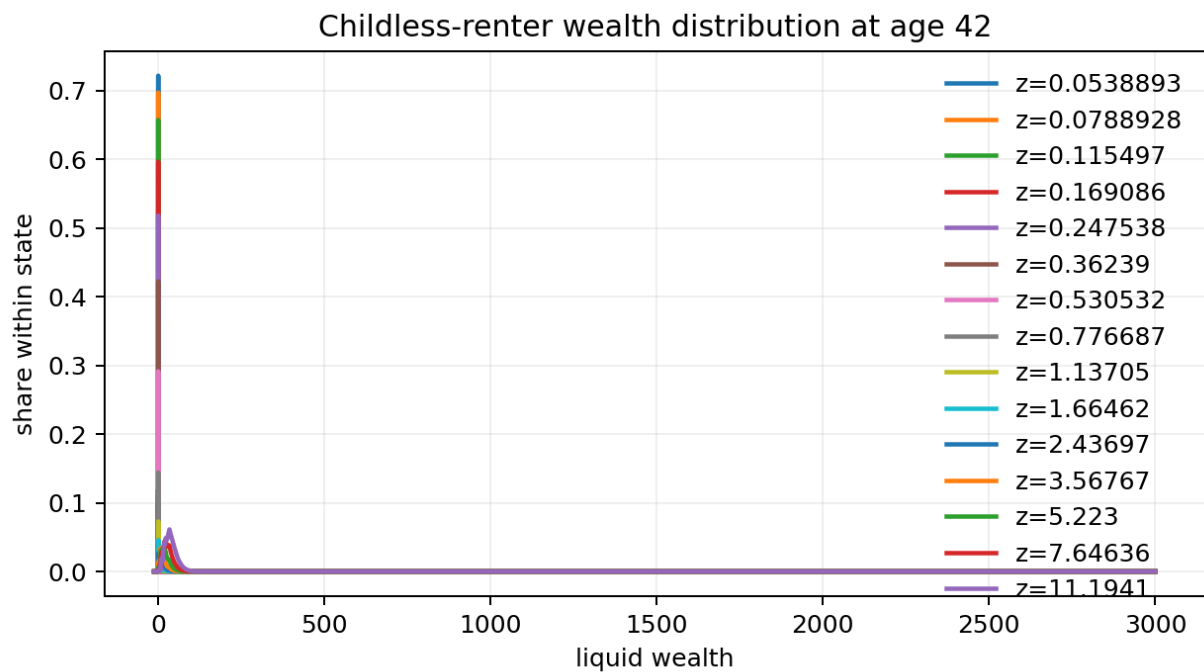


tenure services



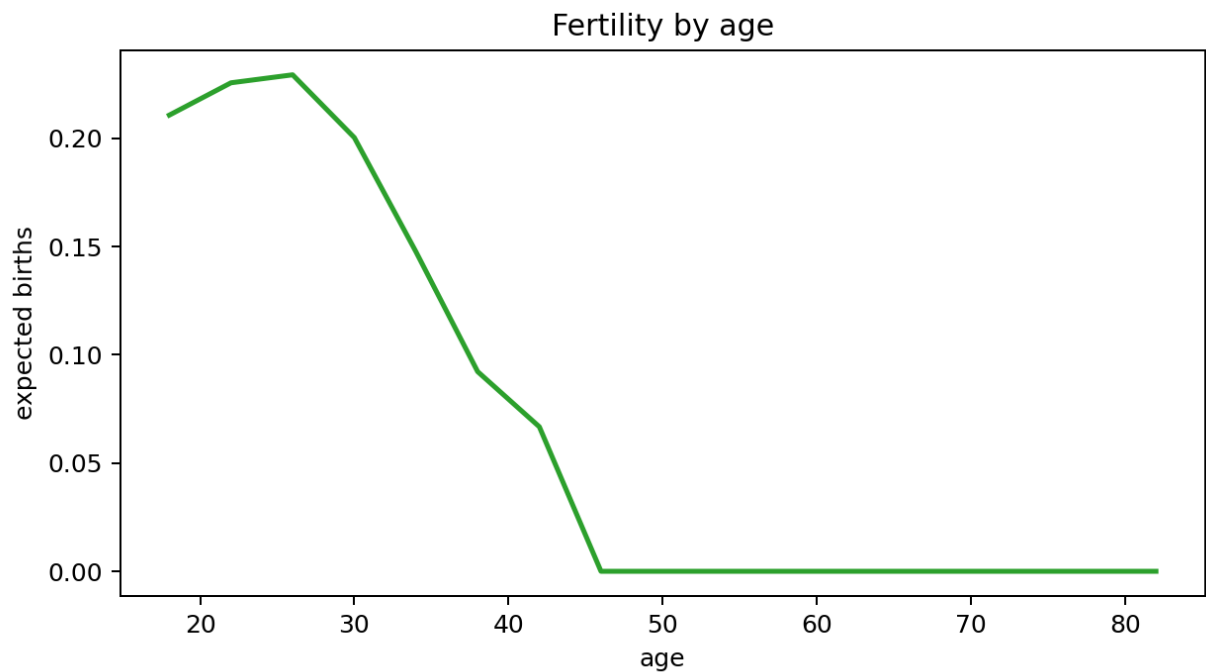
wealth dist childless renter age30

Reference: standard diagnostics

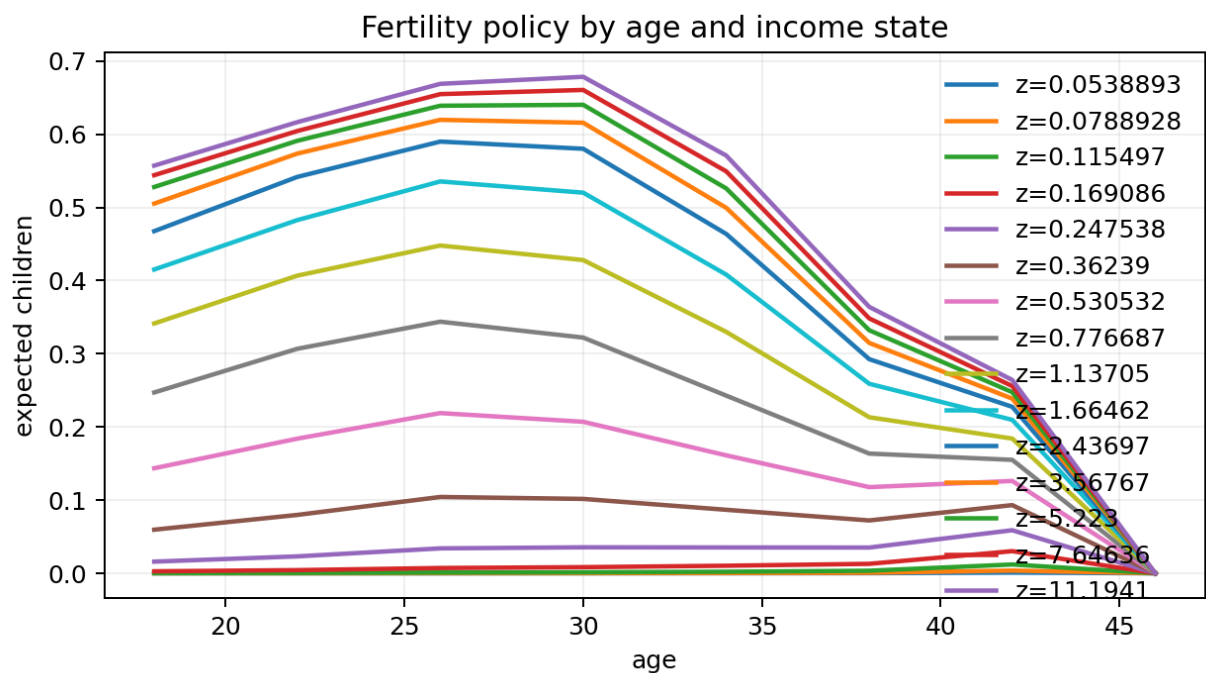


wealth dist childless renter age42

Experimental +10% prescribed price and rent: standard diagnostics

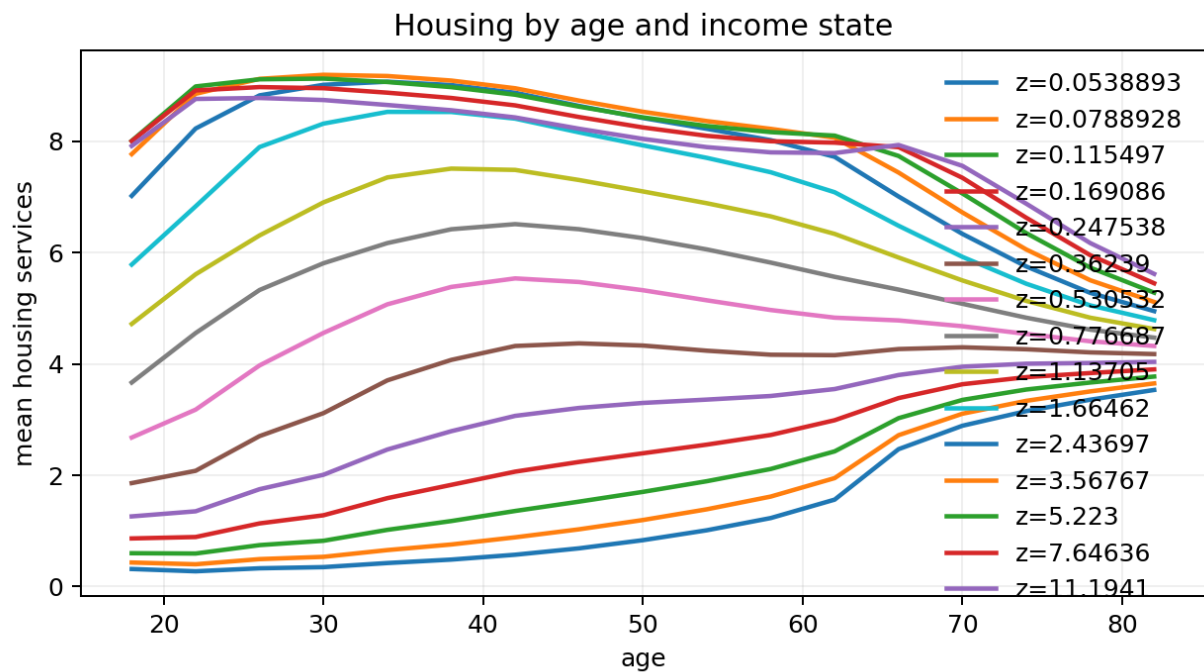


fertility by age

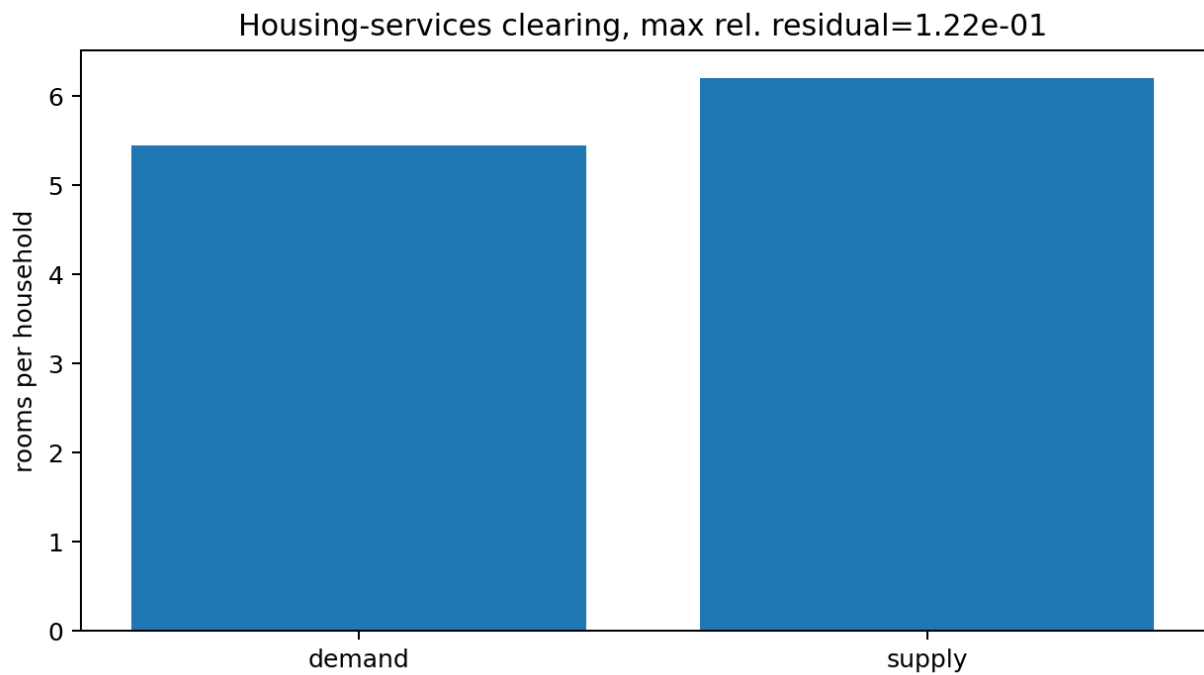


fertility policy by age income state

Experimental +10% prescribed price and rent: standard diagnostics

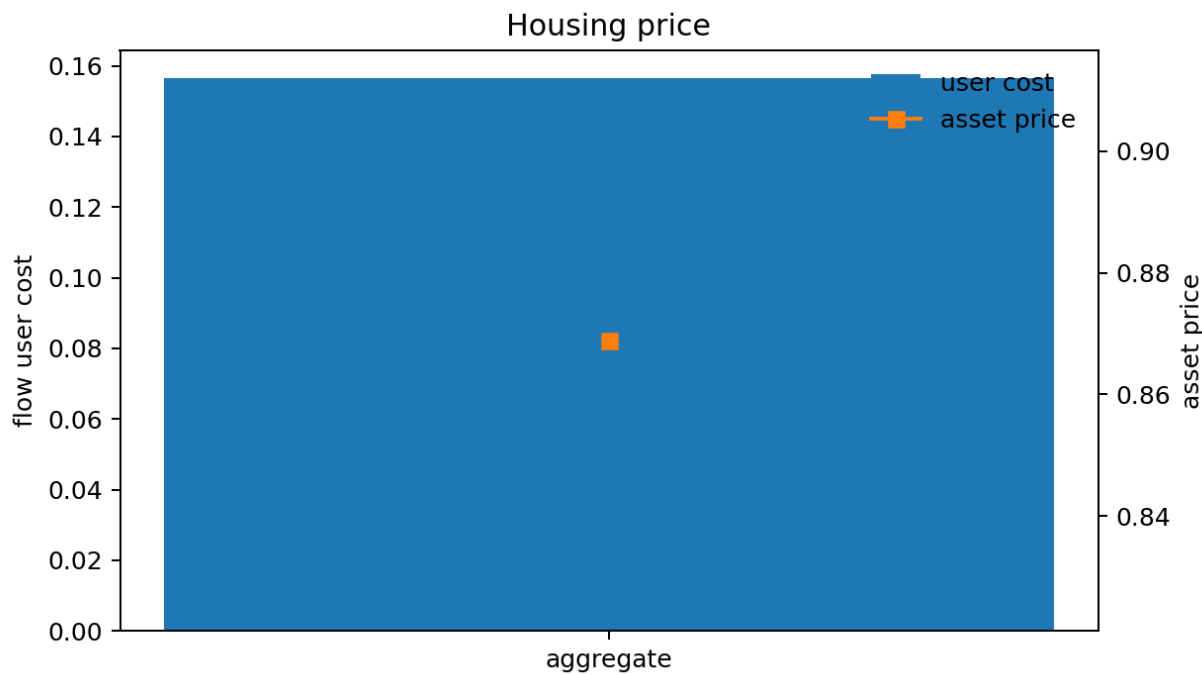


housing by age income state

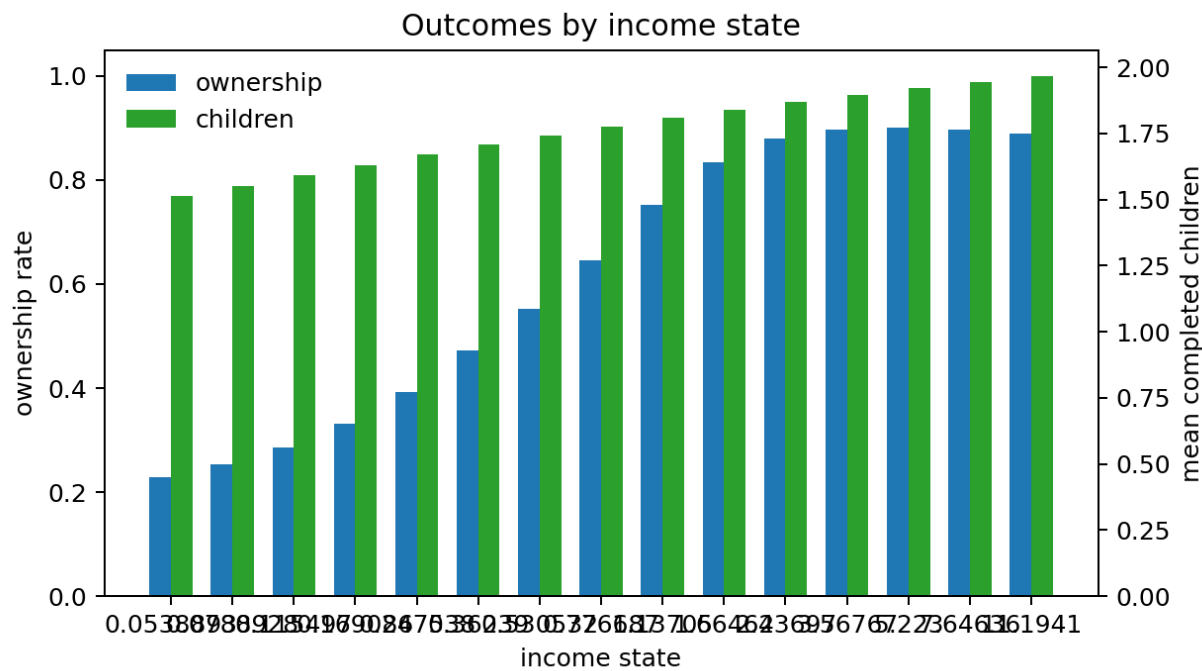


housing market

Experimental +10% prescribed price and rent: standard diagnostics

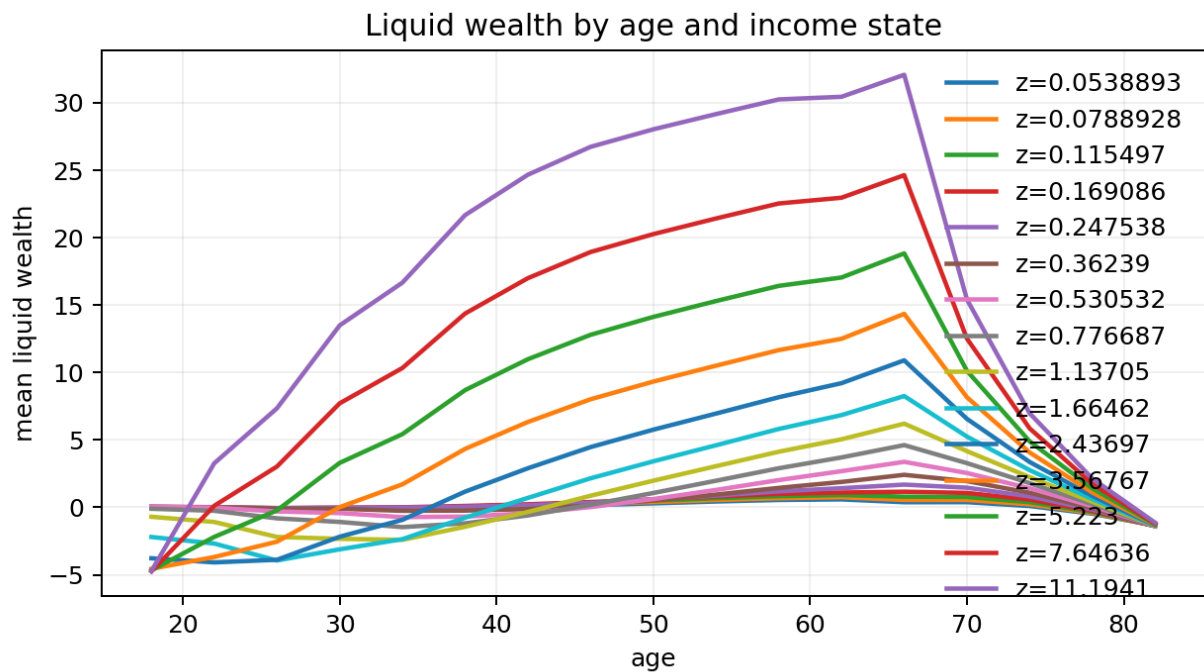


housing prices

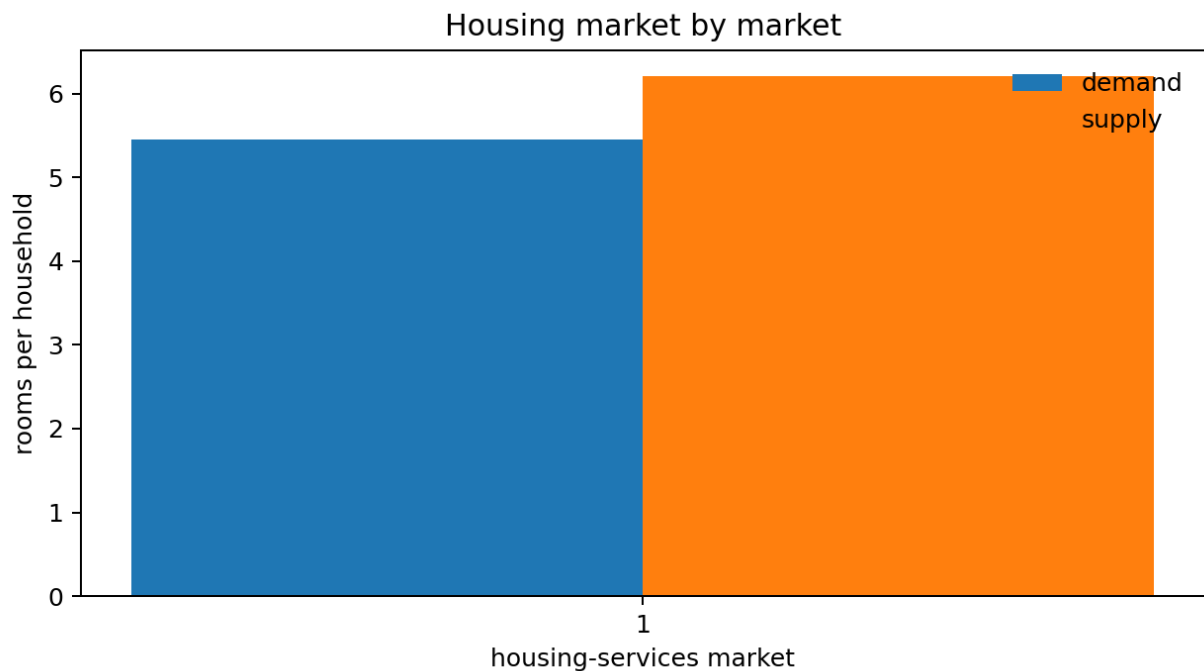


income state outcomes

Experimental +10% prescribed price and rent: standard diagnostics

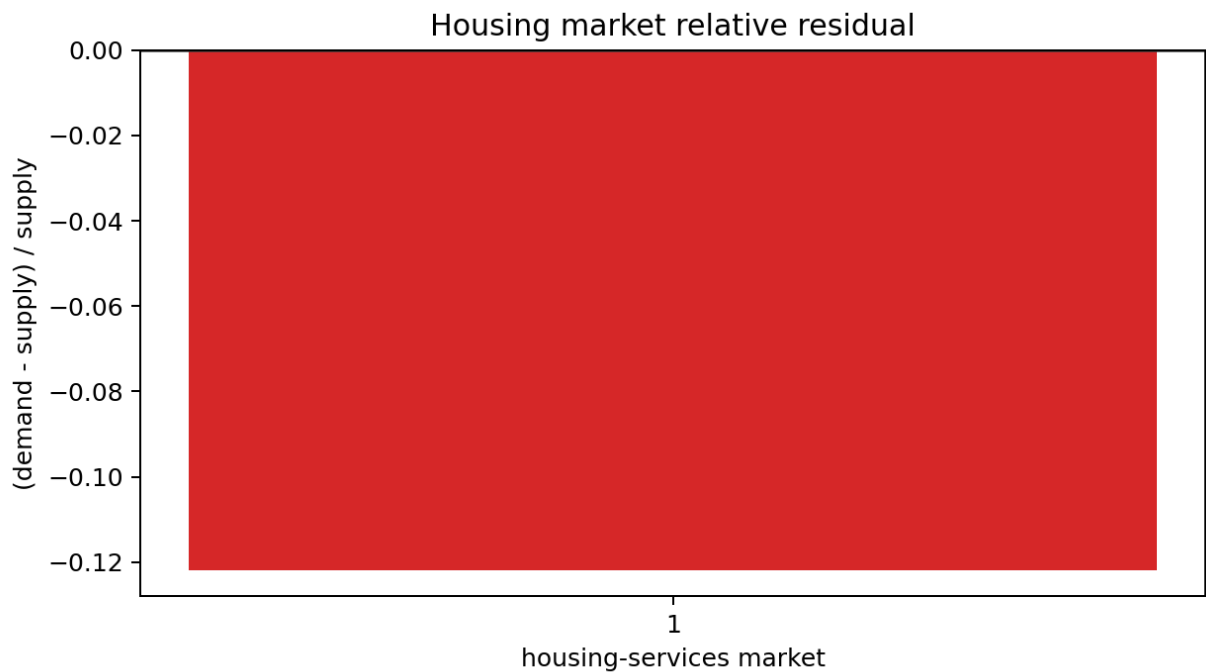


liquid wealth by age income state

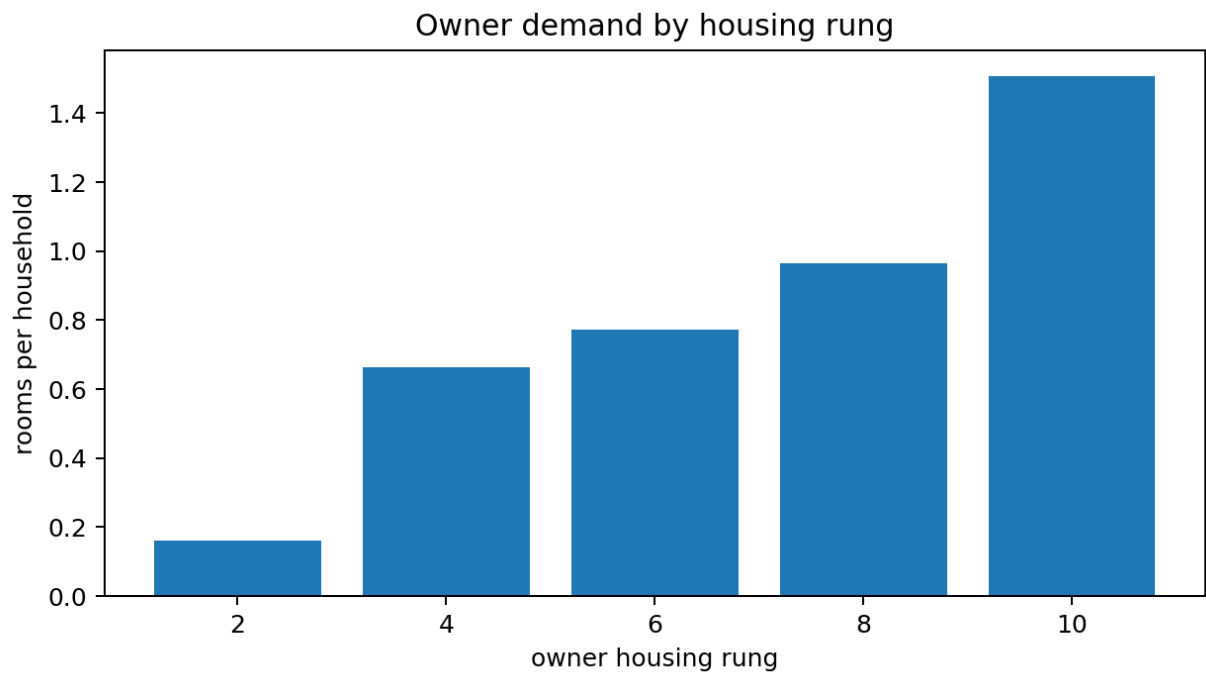


market clearing by market

Experimental +10% prescribed price and rent: standard diagnostics

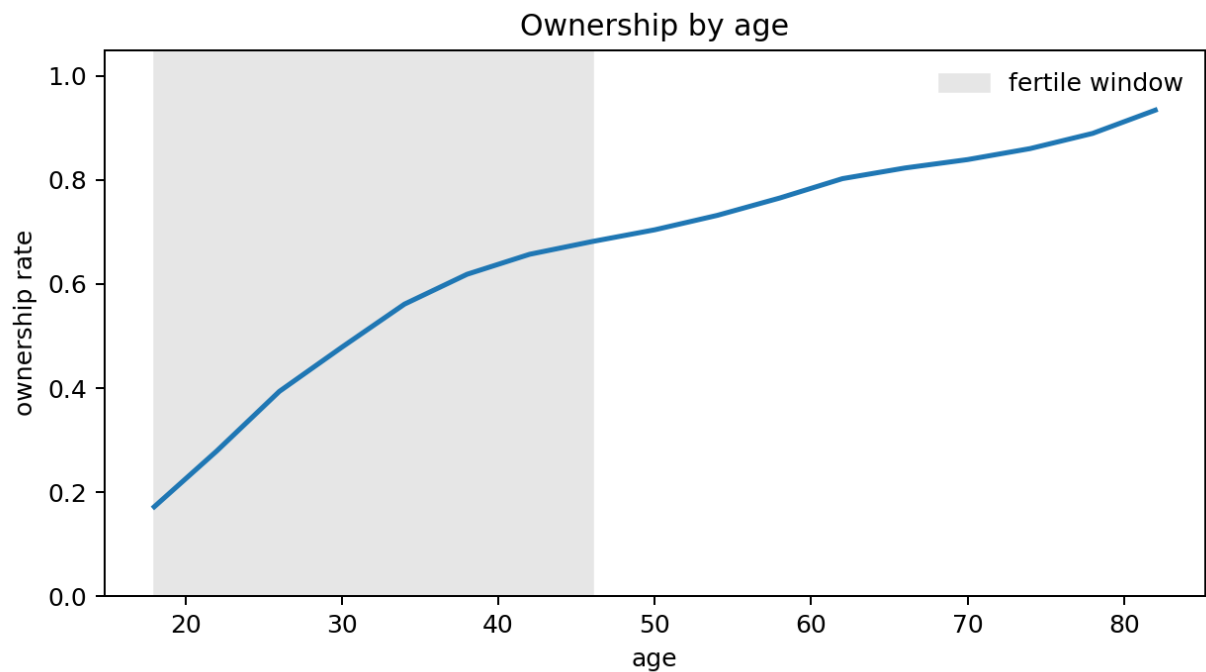


market clearing residuals

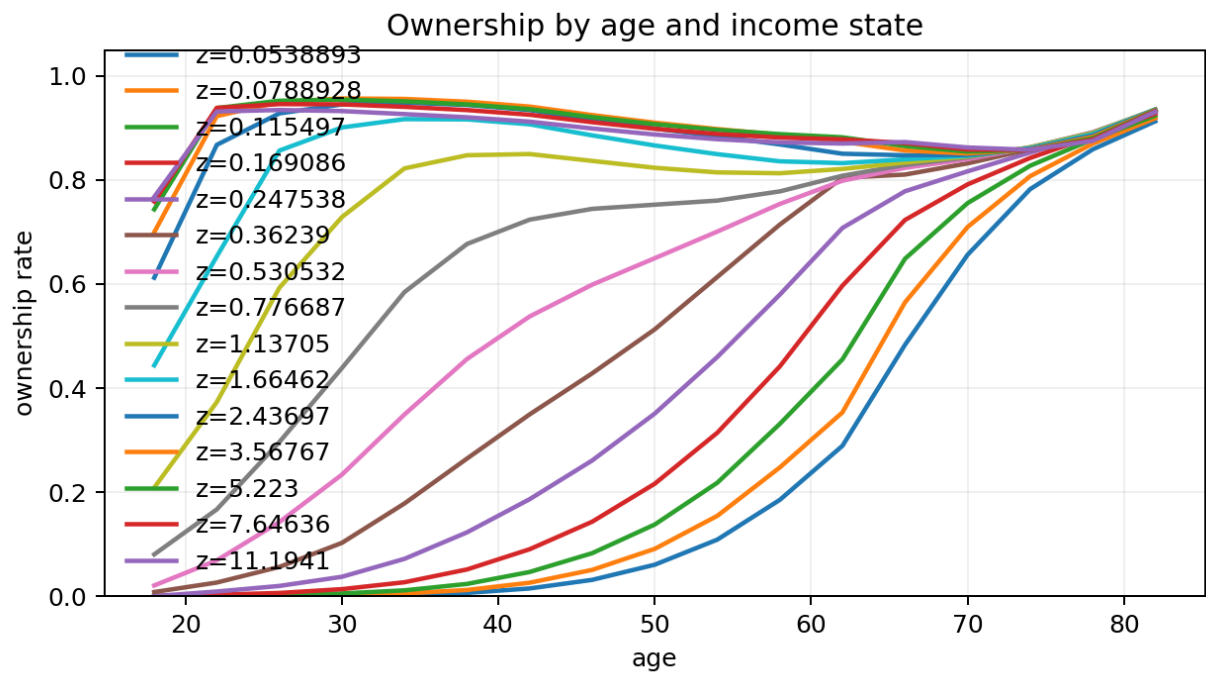


owner rungs

Experimental +10% prescribed price and rent: standard diagnostics

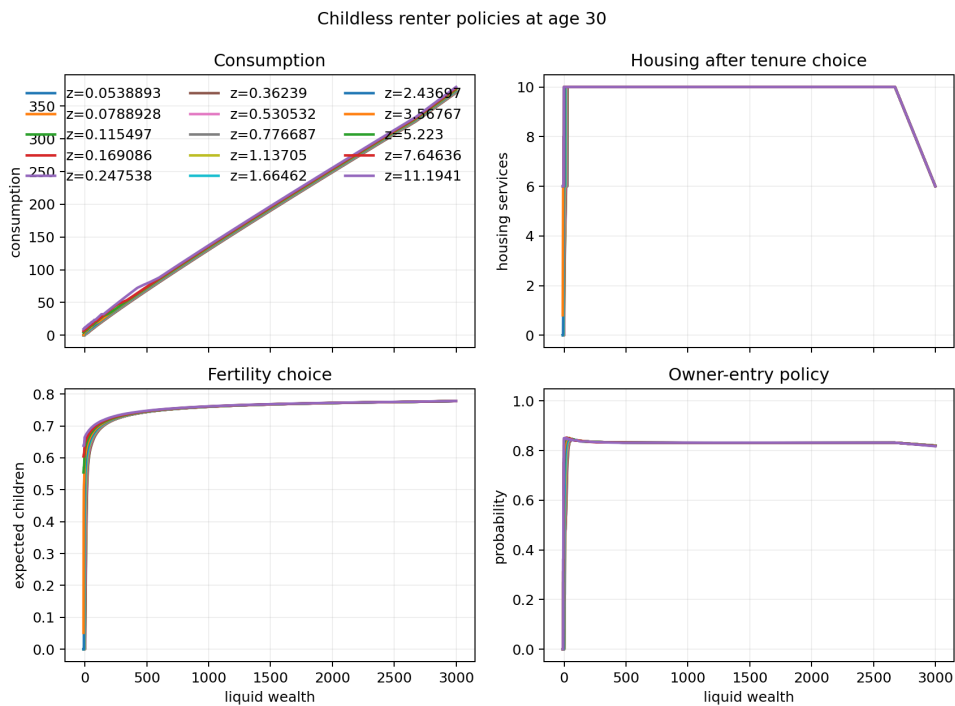


ownership by age

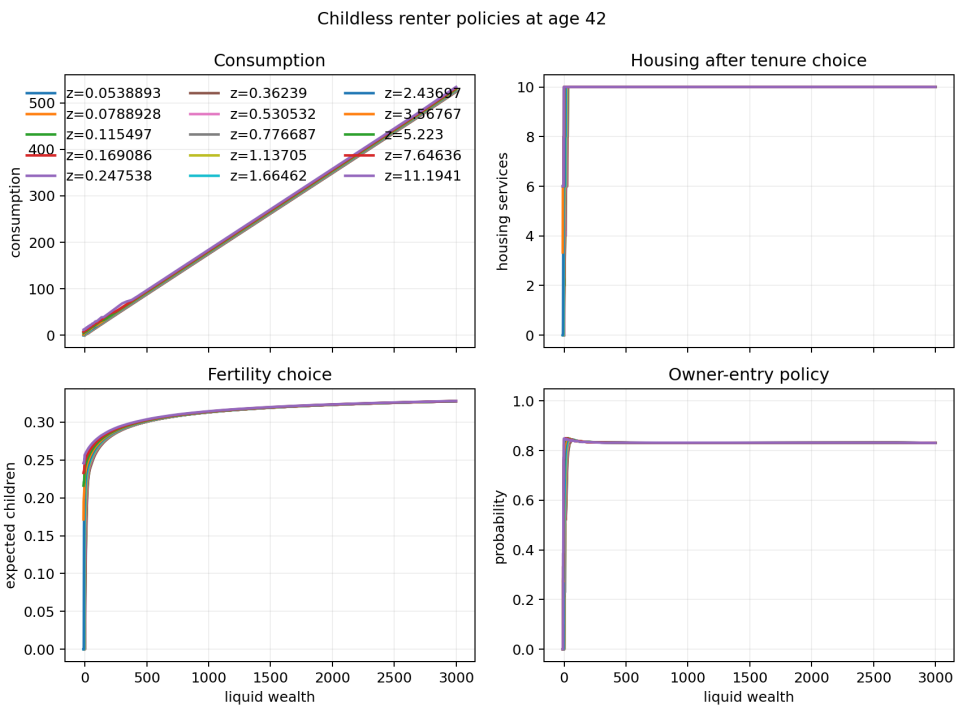


ownership by age income state

Experimental +10% prescribed price and rent: standard diagnostics

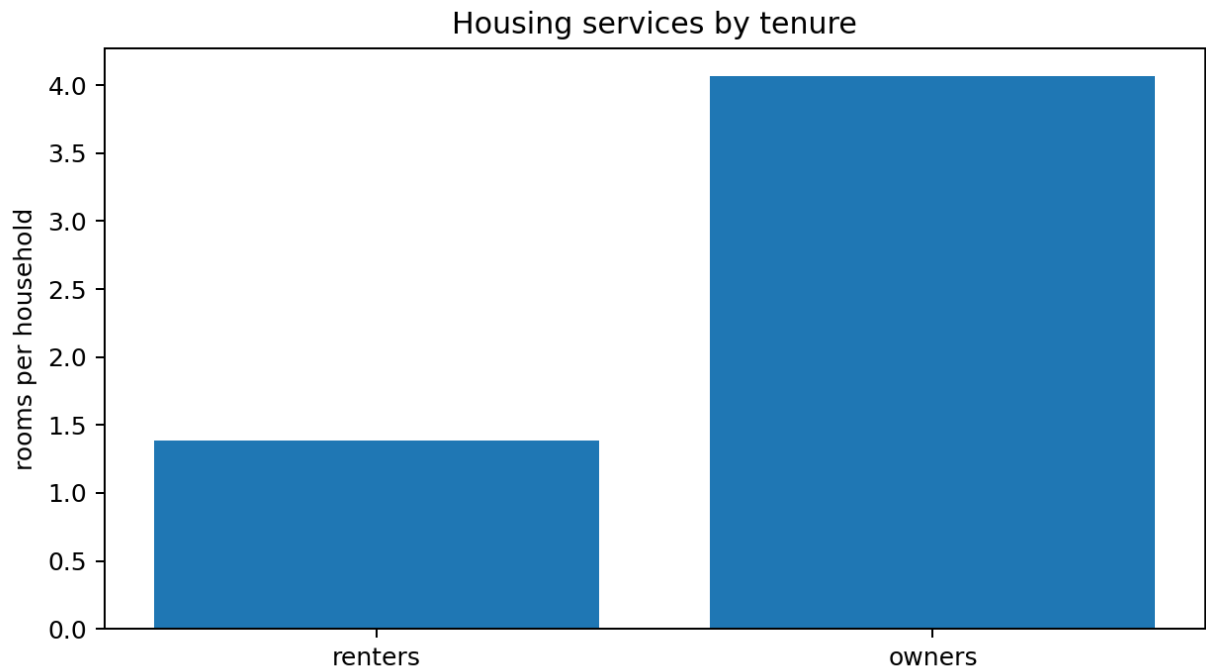


policy childless renter age30

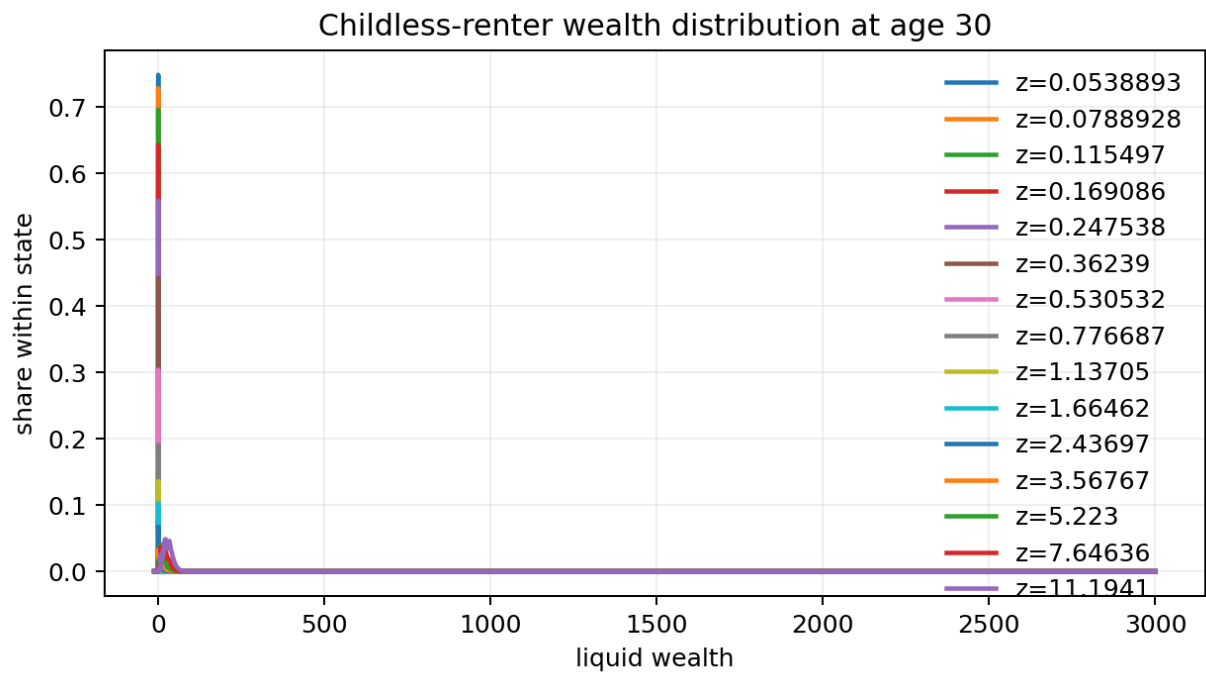


policy childless renter age42

Experimental +10% prescribed price and rent: standard diagnostics

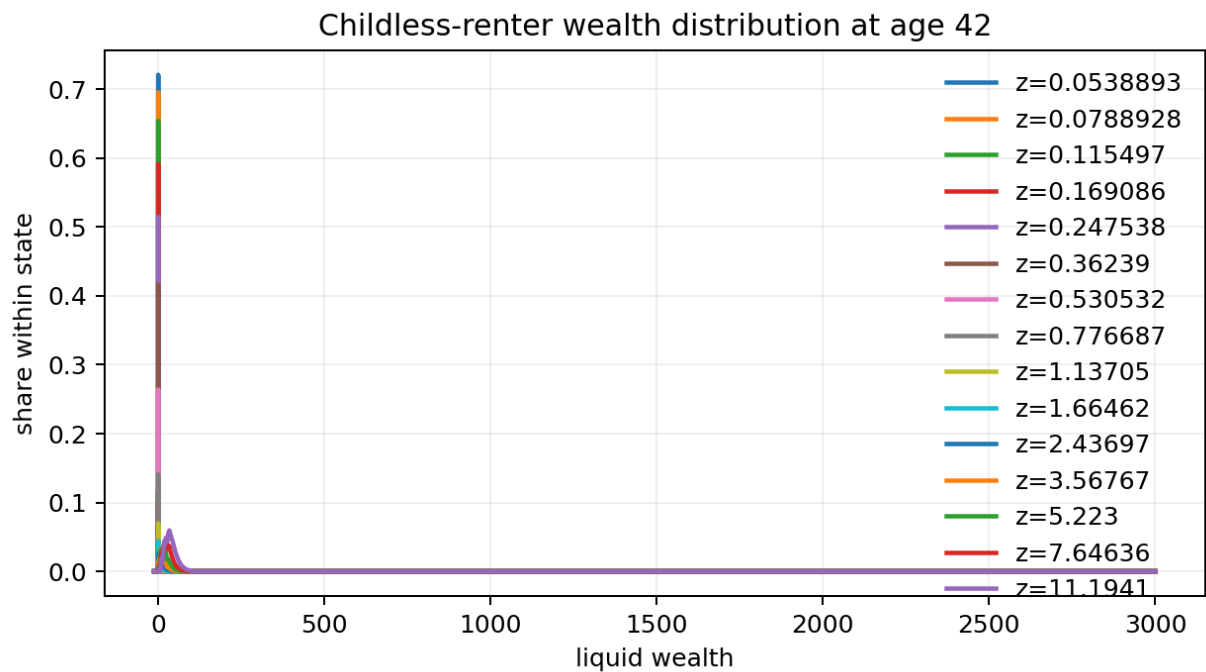


tenure services



wealth dist childless renter age30

Experimental +10% prescribed price and rent: standard diagnostics



wealth dist childless renter age42